

Course: Digital Transformation

Part II

Chapter 4. Common Benefits and Goals (Why Undergo a Digital Transformation?)

Pr Habib Fathallah, PhD ECE



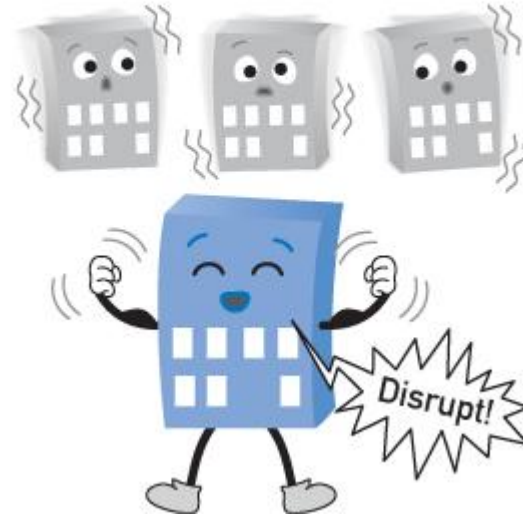
[A Field Guide to Digital Transformation, Roger Stoffers, Thomas Erl.](#)
[Published by Addison-Wesley Professional, Digital Transformation](#)

Pr Habib Fathallah, University of Carthage, Tunisia, TextBook: RogerThomas

Chapter 4. Common Benefits and Goals (Why Undergo a Digital Transformation?)

- Enhanced Business Alignment
- Enhanced Automation and Productivity
- Enhanced Data Intelligence and Decision-Making
- Improved Customer Experience and Customer Confidence
- Improved Organizational Agility
- Improved Ability to Attain Market Growth

Chapter 4. Common Benefits and Goals (Why Undergo a Digital Transformation?)



- *Now comes the fun part. We get to learn about the many positive things a digital transformation can bring. Organizations undergoing digital transformation often look forward to “shaking things up” in their markets and communities.*

Enhanced Business Alignment

Traditionally, organizations were often structured around business silos based on specific products, services or lines of business.

A digital transformation can introduce the need for:

- previously isolated or separated business departments to collaborate in support of common business goals ([Figure 4.1](#))

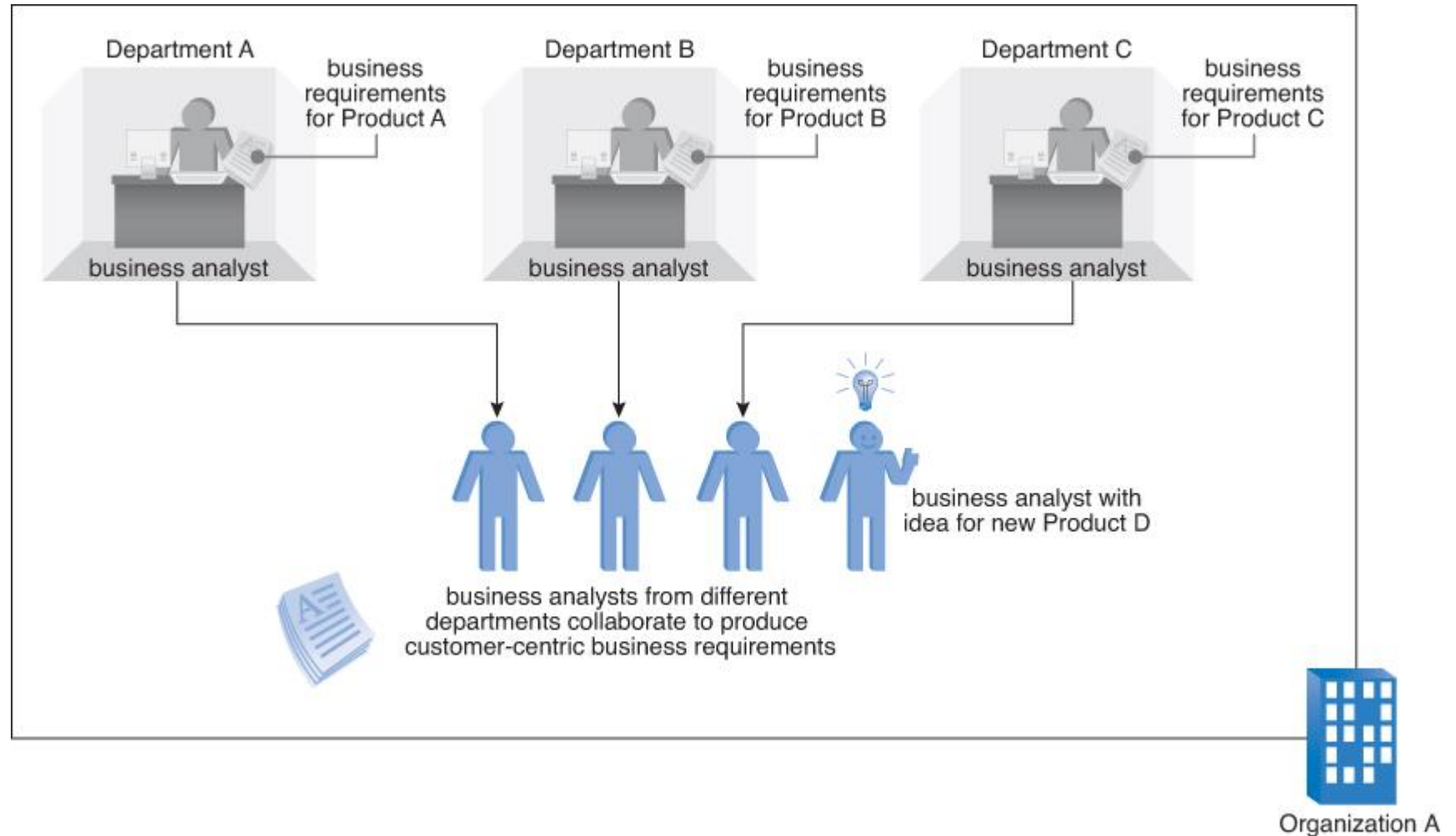
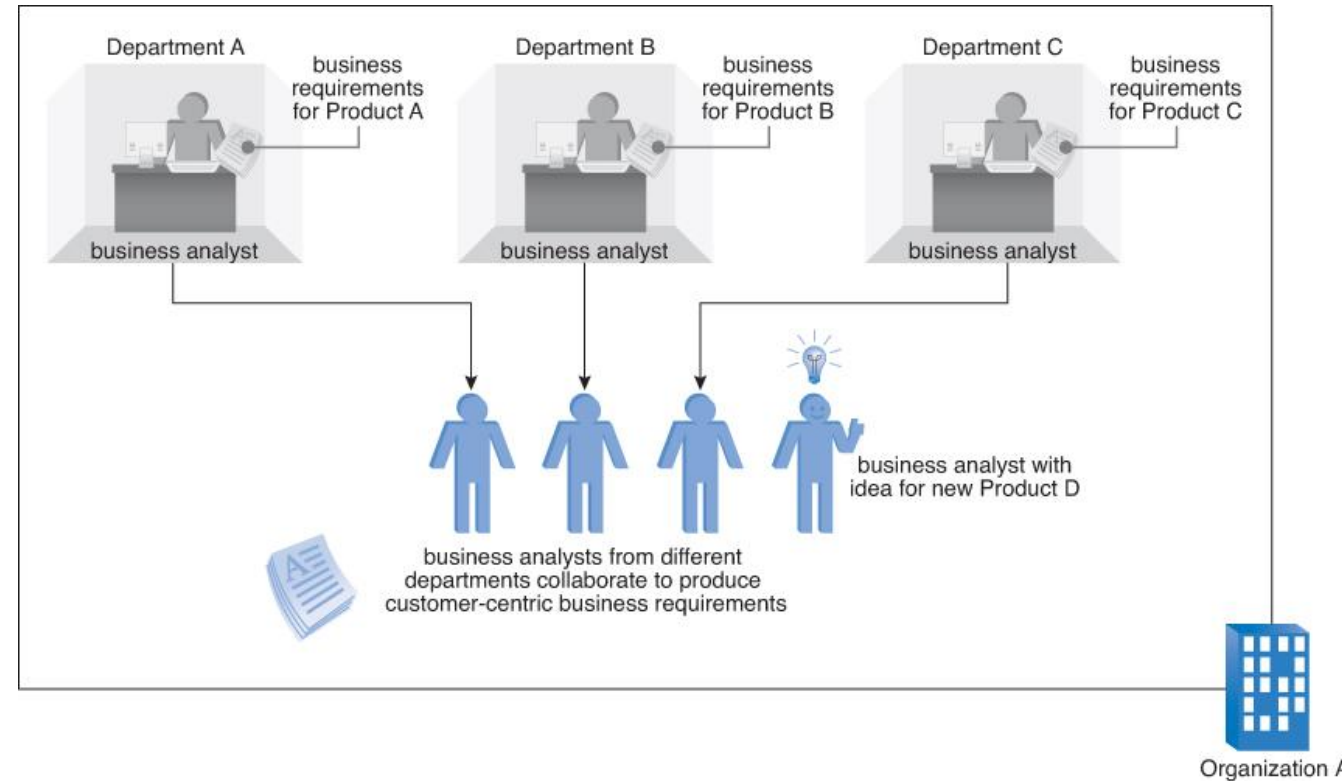


Figure 4.1

A common goal of digital transformation initiatives is **to eliminate product “silos”** so as to establish an environment that fosters collaboration and alignment across departments.

For example, to **improve customer-centricity**, those groups or departments originally responsible for business analysis as it pertained to individual products, now work together to provide a consolidated customer experience through which all products (and new products) can be explored.

New, broader performance and customer success metrics and indicators are commonly established to measure the collective outcome of these types of collaborations instead of measuring only the performance of individual contributions.



- previously separated business and IT departments to collaborate more closely in support of common business goals
- existing business processes and models to be optimized, reengineered and/or further innovated in support of new business goals
- single-purpose business processes previously focused on specific products to be consolidated with others in support of new business goals ([Figure 4.2](#))
- new business processes and models to be introduced and merged with existing business processes and models in support of new business goals ([Figures 4.1](#) and [4.2](#))

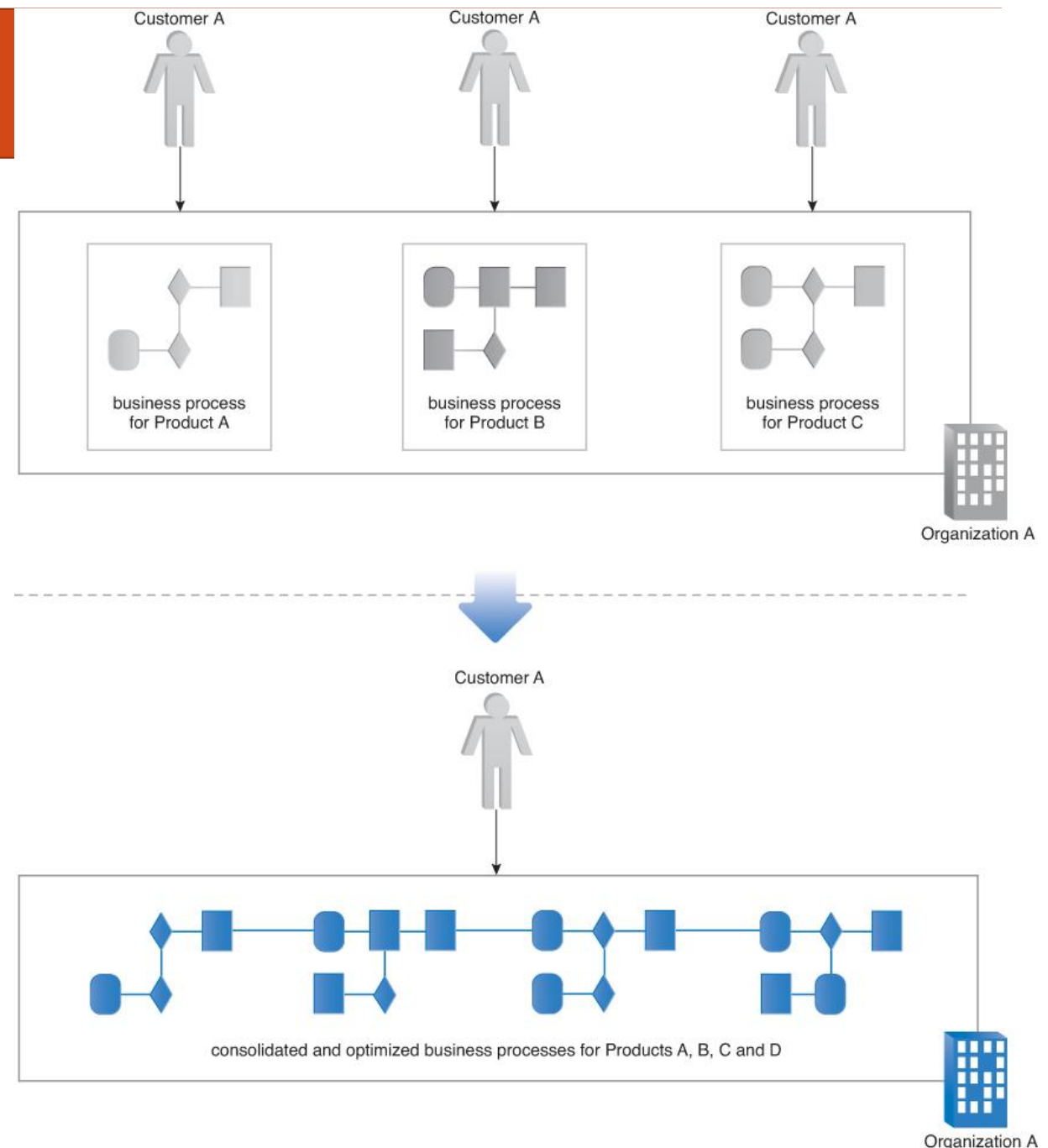




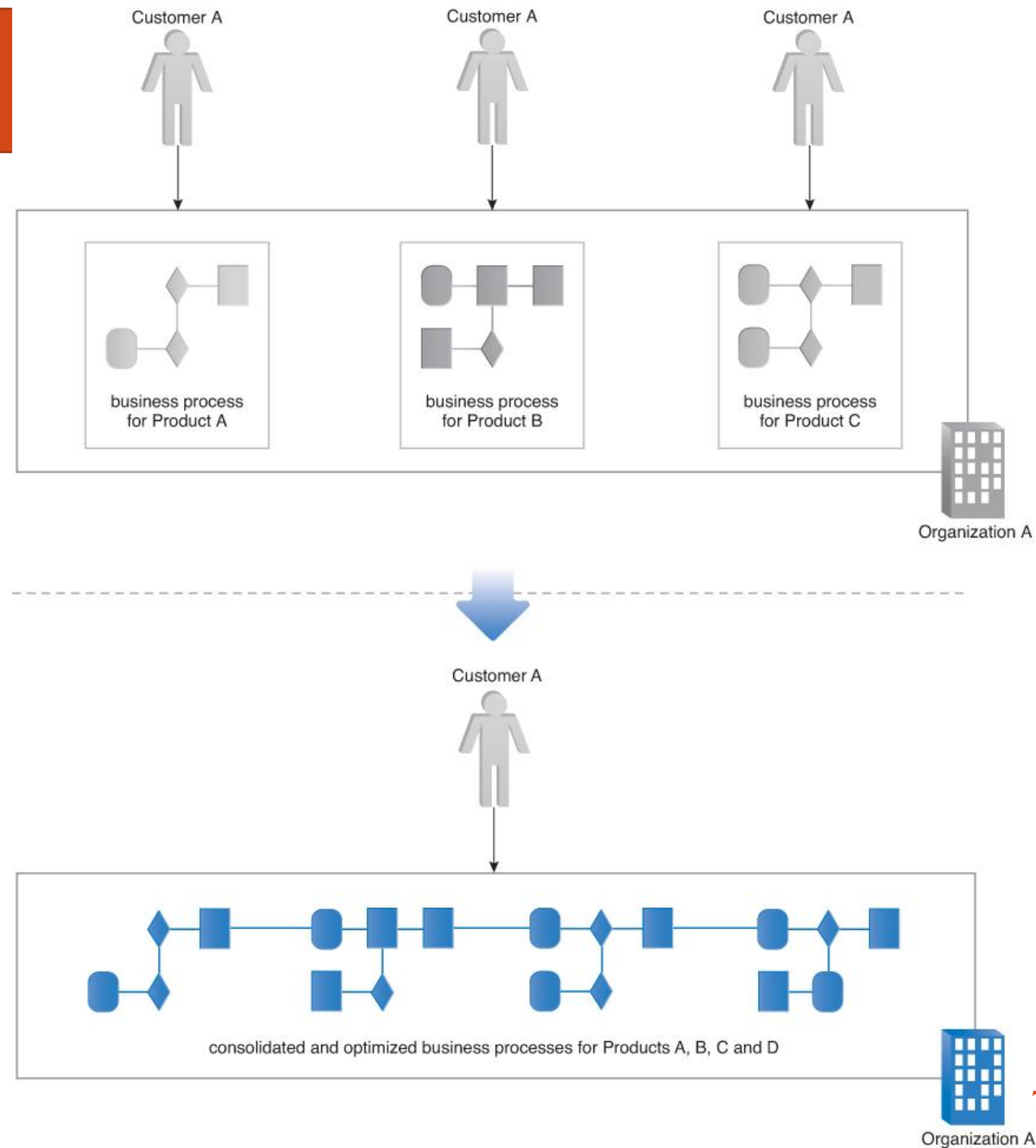
Figure 4.2

Customer A wants to obtain three different products from Organization A.

Previously, Customer A had to interact with Organization A via three separate workflows and systems (top), which may have even required the creation of three individual accounts.

A transition toward a customer-centric solution results in a consolidated customer experience enabling Customer A to carry out transactions in relation to the three products in a single environment (bottom).

Customer A is further able to discover new products while in the consolidated environment.





These business transformations and the resulting cross-departmental collaborations that are formed naturally align the business of an organization with its strategic business goals, several of which may be focused on improving customer-centricity.

This type of business alignment can strengthen an organization culturally, but primarily benefits the organization by establishing a solid foundation upon which automation and data science technology enhancements can be applied.

These technologies can be effectively utilized by human workers to enable the organization to realize its business goals to their full potential.

Enhanced Automation and Productivity

An organization undergoing a digital transformation can extend the reach and improve the quality of its automation capabilities significantly. Solutions can be built using combinations of technologies that can enable organizations to automate business tasks so as to boost operational productivity.

For example, automation technologies can:

- ☐ automate tasks that previously needed to be performed manually
- ☐ automate new tasks in support of new products and services
- ☐ automate data collection across environments outside of organization boundaries
- ☐ automate actuating tasks in remote devices outside of organization boundaries
- ☐ automate tasks at a higher usage capacity than what was previously achievable
- ☐ automate tasks more reliably than what was previously possible
- ☐ automate decision-making
- ☐ improve the security and quality of storage for private, sensitive and important business data

What is significant about digital transformation environments is how they combine these technologies into distinct platforms that help achieve strategic goals via their collective features ([Figure 4.3](#)).

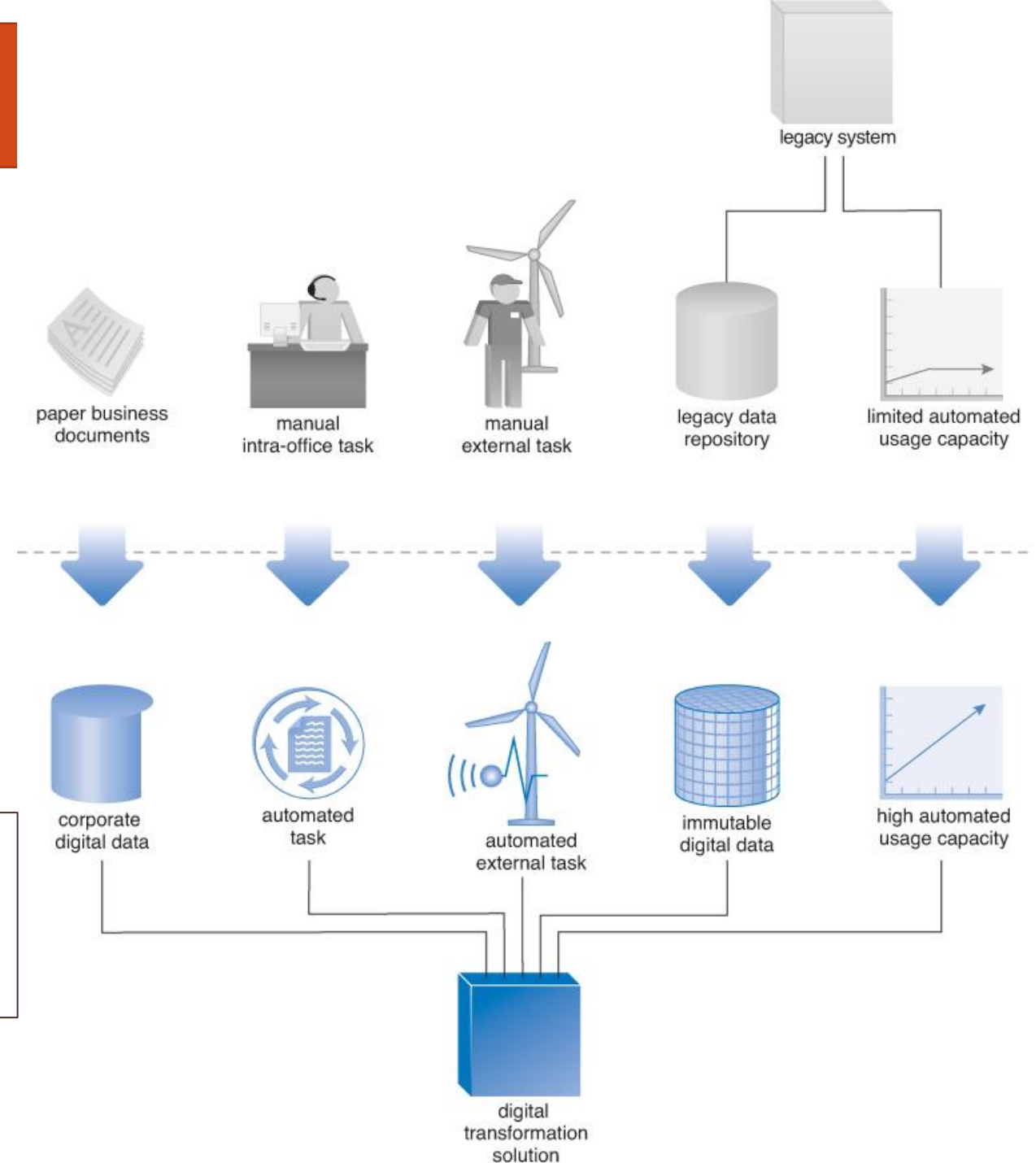


Figure 4.3

The application of digital transformation technology improves the quality and efficiency of a range of operational business tasks.

Enhanced Data Intelligence and Decision-Making

As previously explained, digital transformation solutions can accumulate valuable data intelligence, enabling them to produce deeply insightful analysis results in realtime or near-realtime.

This can significantly empower organizations with new insights, new ideas and more decisive and successful decision-making capabilities.

Organizations further have the option to defer some decision-making responsibilities to the underlying digital transformation solutions themselves.

When doing so, decisions can be made and executed at the same rate (realtime or near-realtime) as that of the data processing ([Figure 4.4](#)).

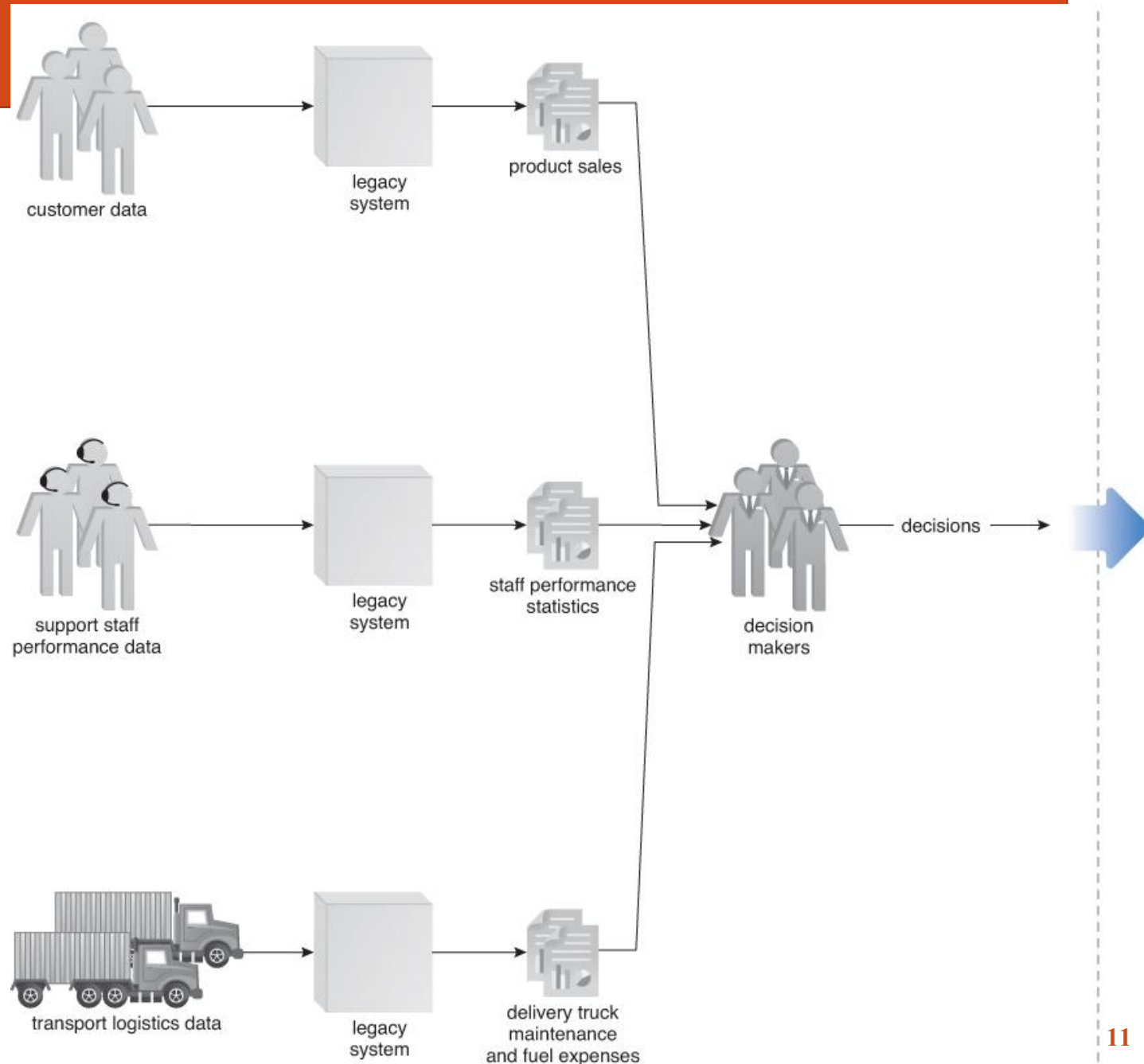


Figure 4.4

Traditional legacy systems produce various independent reports for human decision makers.

Digital transformation solutions process and consolidate input data from a range of sources with the aim of producing enhanced reports at a faster rate and with greater data intelligence.

The reported data may be provided to human decision makers or to data science systems that can make and act upon decisions autonomously.

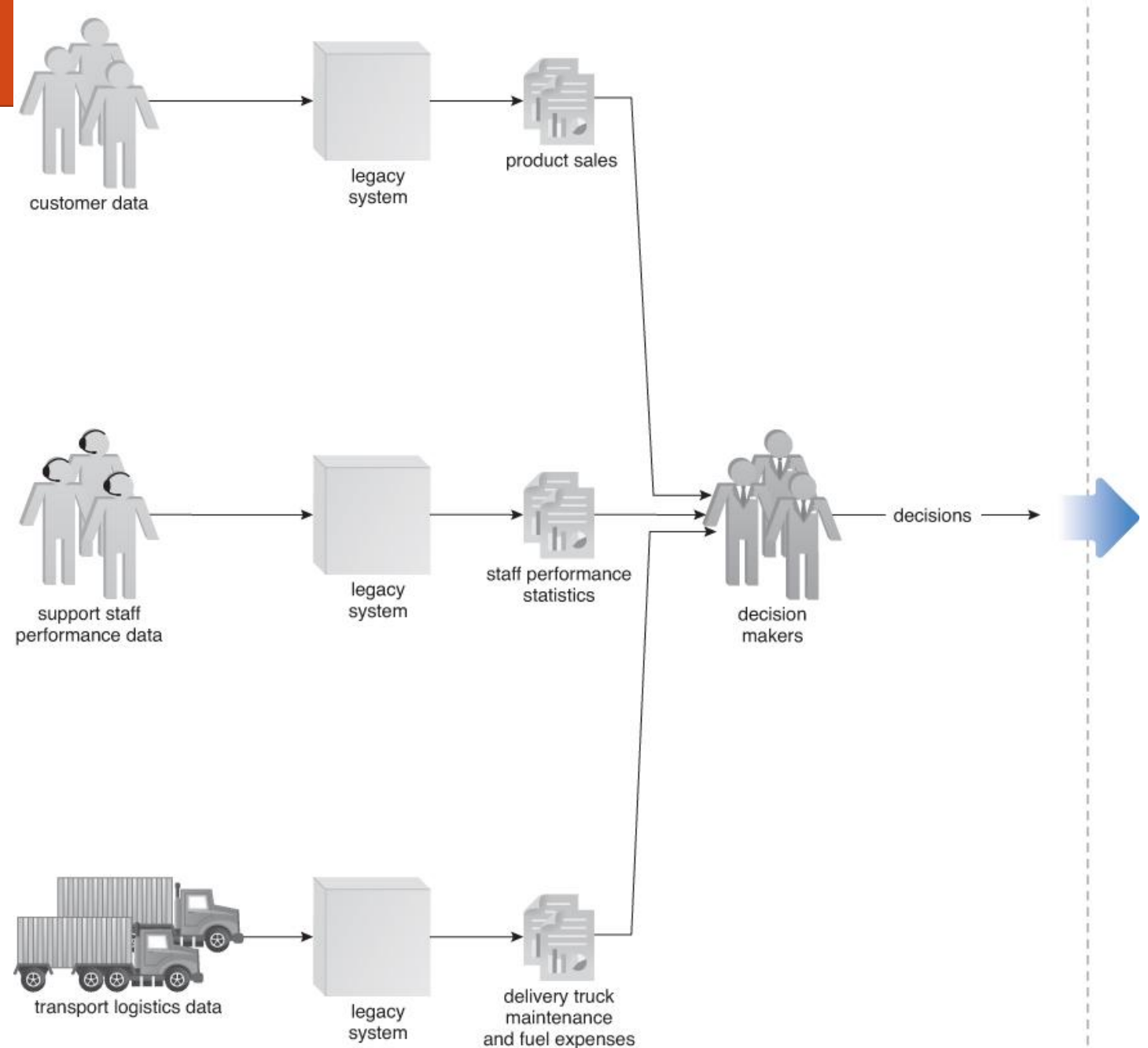
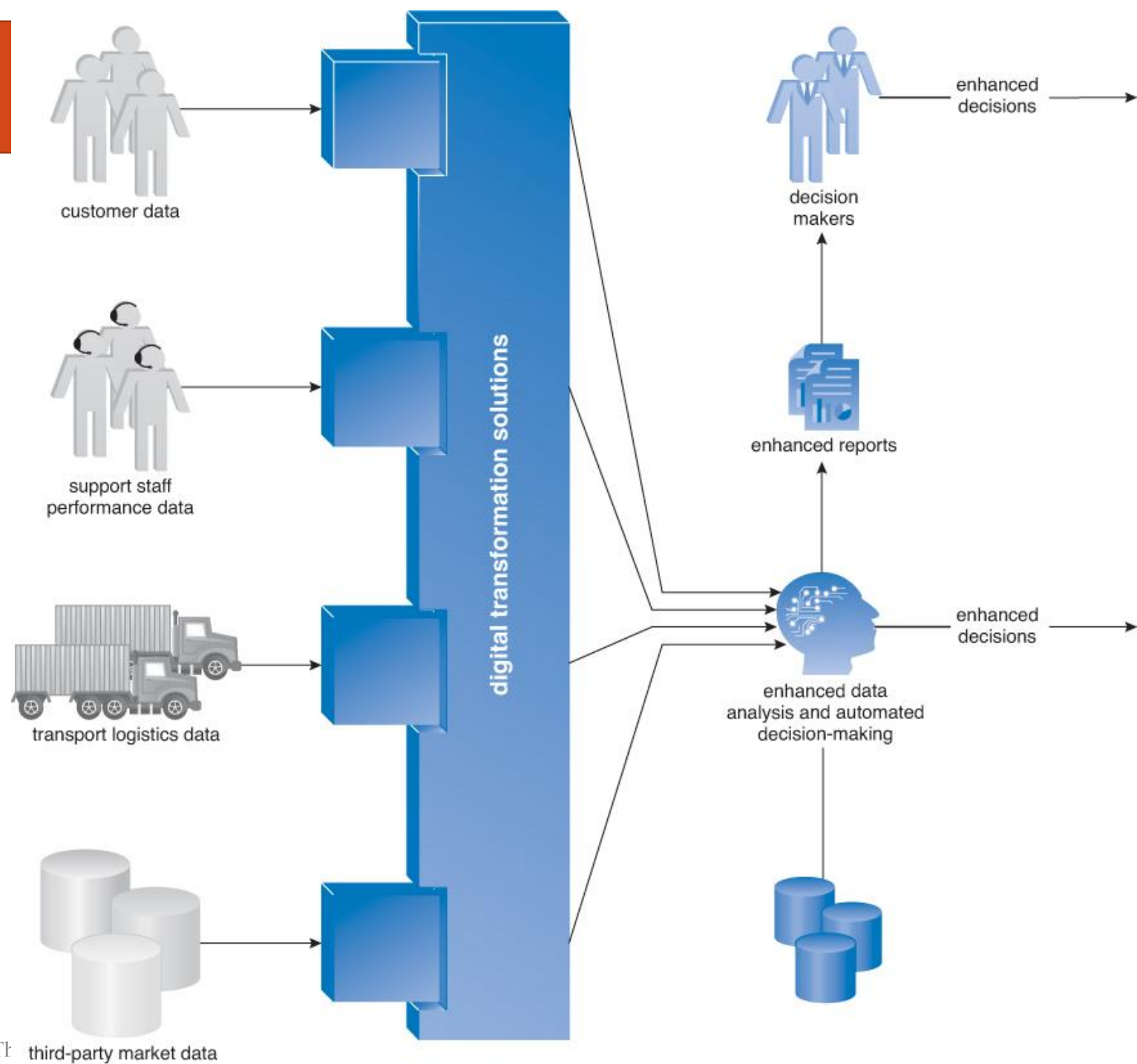


Figure 4.4

Traditional legacy systems produce various independent reports for human decision makers.

Digital transformation solutions process and consolidate input data from a range of sources with the aim of producing enhanced reports at a faster rate and with greater data intelligence.

The reported data may be provided to human decision makers or to data science systems that can make and act upon decisions autonomously.



Improved Customer Experience and Customer Confidence

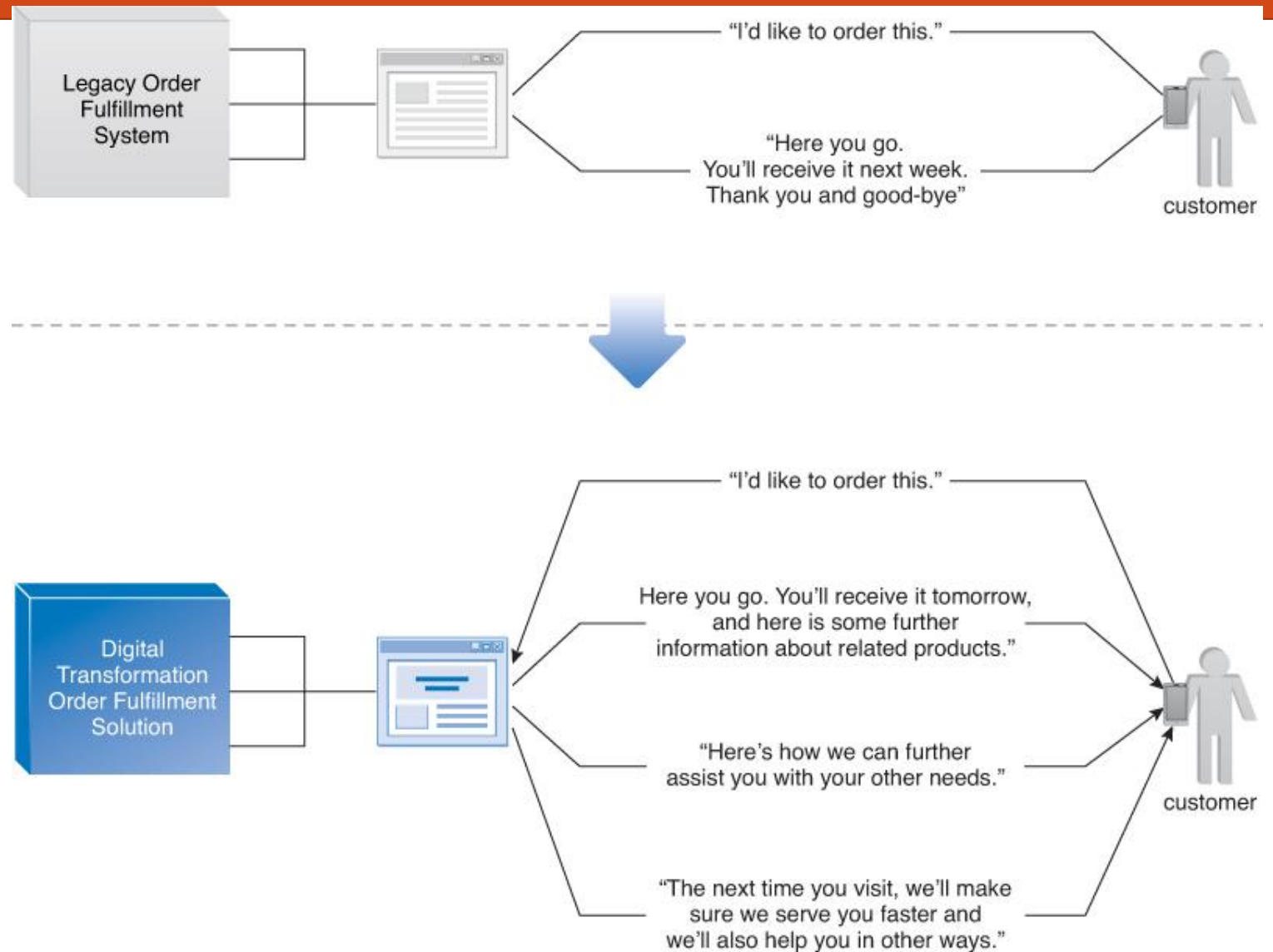
One of the foundational objectives of digital transformation is to foster a shift toward establishing a customer-centric culture, resulting in improved relationships with customers, attracting new customers and supporting all of this via enhanced automation.

Customer-centric solutions have the potential of capturing the interest and enhancing the satisfaction and confidence of customers.

This brings with it several core benefits, including:

- increasing the speed at which customers are served by reducing the time-to-value of services
- improving the effectiveness with which services are delivered to customers by enhancing their quality
- increasing the “warmth” of the customer experience
- improving customer confidence and loyalty by maintaining on-going relationships with customers beyond individual transactions

Digital transformation solutions aim to achieve these improvements by being designed, from the ground up, with customer-centricity in mind ([Figure 4.5](#)), as further explored in [Chapter 6: Realizing Customer-Centricity](#).



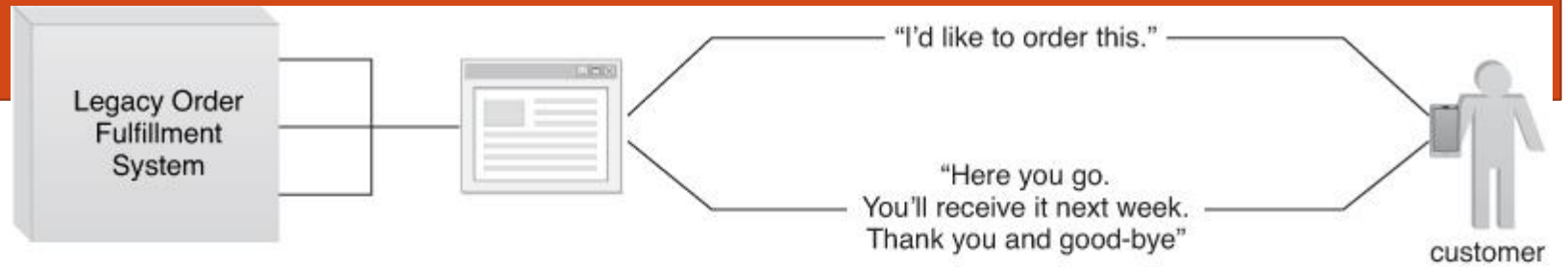
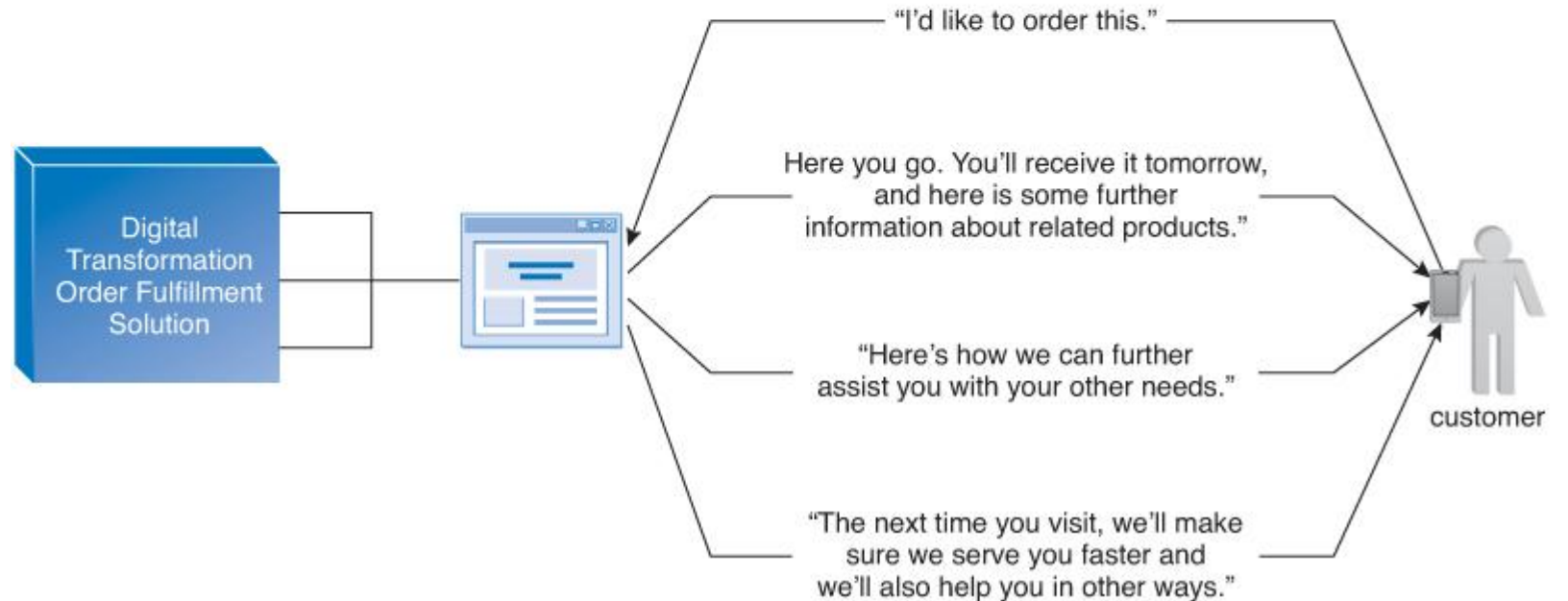


Figure 4.5

Digital transformation solutions are designed to be customer-centric so as to enable customers to interact with an organization in new ways and to make the customers' experiences as positive and effective as possible.





Improved Organizational Agility

Digital transformation can transform an organization to become more agile in its ability to:

- Adapt to unforeseen business changes, such as new or existing disruptive competitors (that may be introducing new or improved products into a market), internal changes (the resignation of key executives, changes in internal funding, labor unrest), regulatory changes (new taxes, policies and other regulations that affect how a business operates) ([Figure 4.6](#)).

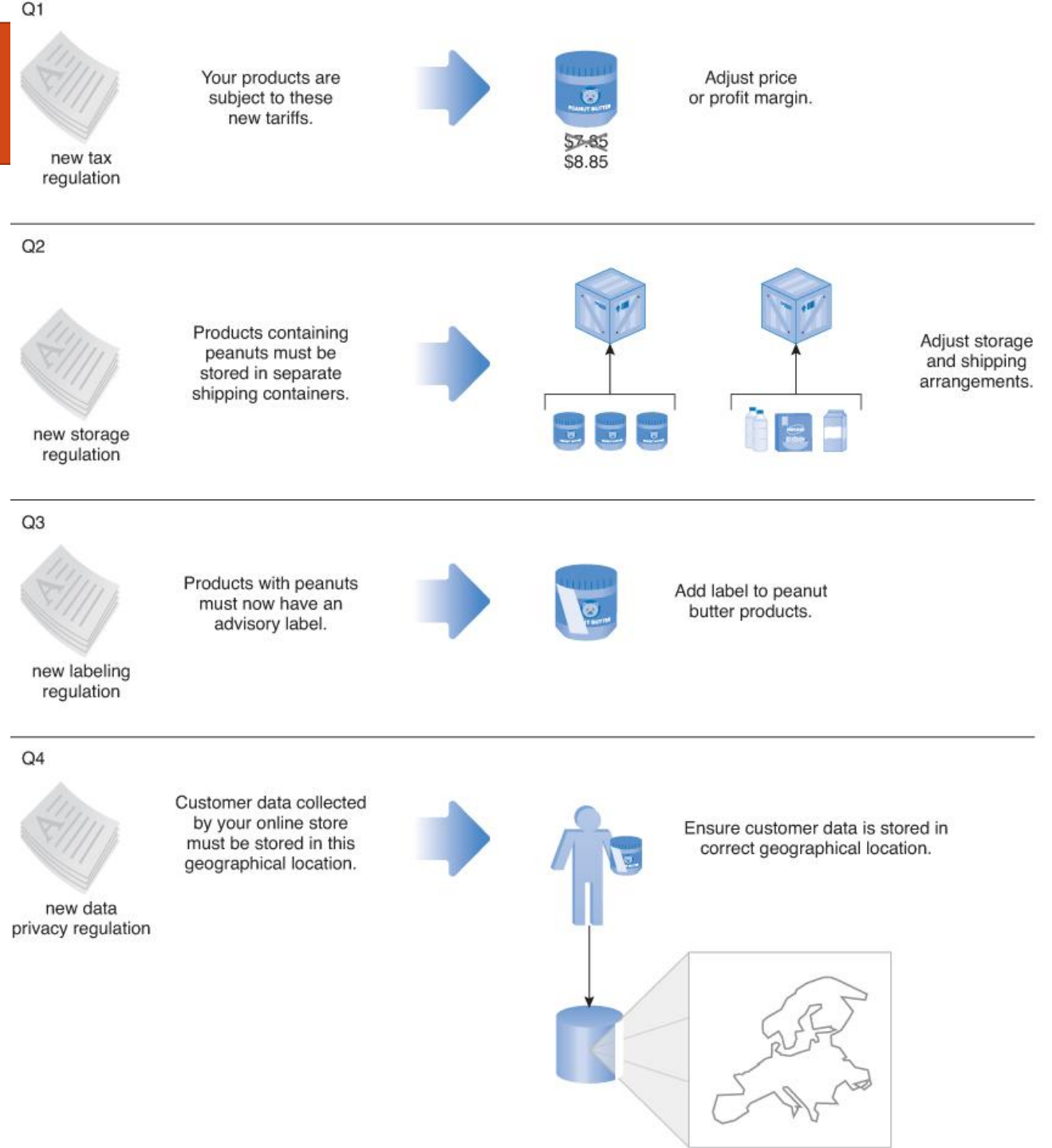
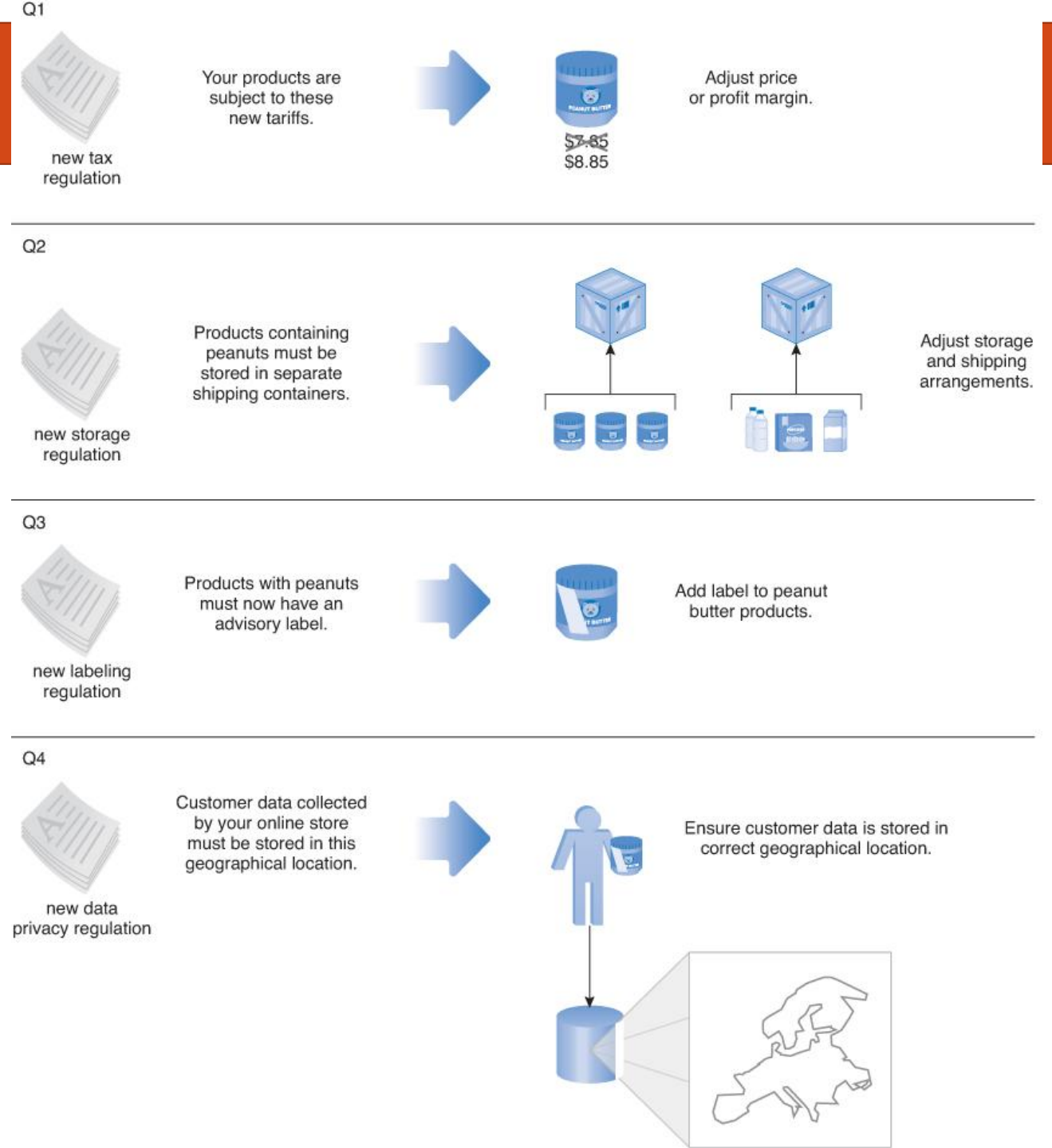


Figure 4.6

Over the course of a year, a business selling food products is required to adapt to a series of regulatory changes that impact some of its products.

The improved alignment of its underlying automation systems allows the organization to more efficiently adjust its operations to accommodate such unforeseen changes.



Improved Ability to Attain Market Growth

Digital transformation platforms enable an organization to make significant enhancements in how its business currently operates and, often more importantly, to introduce new products and services to disrupt existing markets in pursuit of growth.

This can lead to several avenues for increasing market share and revenue, such as:

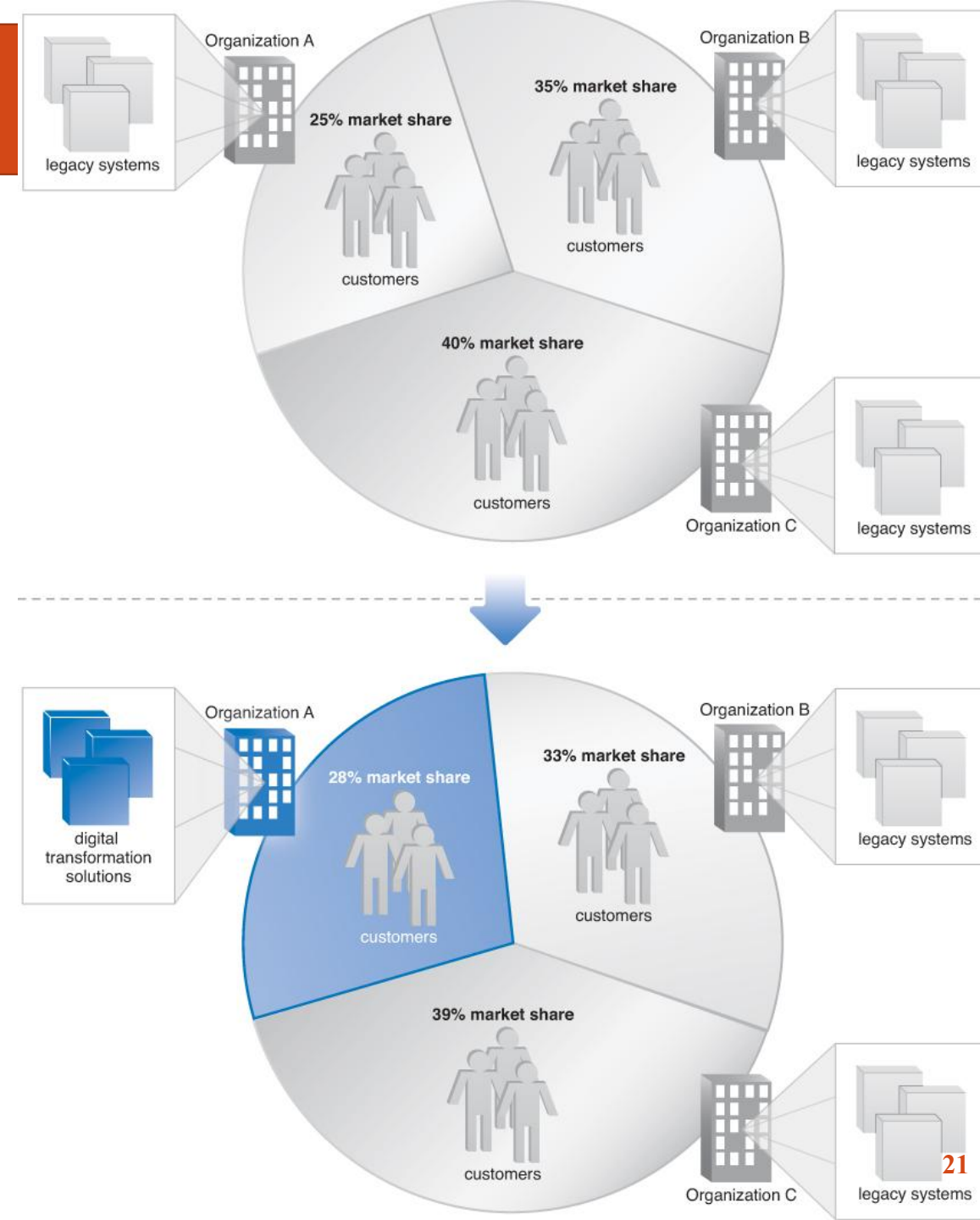
- being able to reach a wider range of customers by moving more products and services online
- being able to reach new customers by adding new products and services to their offerings
- being able to increase the frequency of (existing and new) customers returning by improving customer experience



Furthermore, the technologies associated with digital transformation provide many opportunities for underlying automation solutions to become highly optimized, such as:

- optimizing business workflows by improving the quality of automation technology
- optimizing organization-wide workflows by introducing new automation technology in support of cross-departmental collaboration
- carrying out tasks faster and with less overhead by replacing manual labor with automation logic
- carrying out decisions in realtime and with less expense by replacing human decision makers with automated decision logic
- continually improving and refining business operations in response to new digital data intelligence that is collected, analyzed and fed into decision-making responsibilities carried out by humans and machines

By repeatedly building upon these enhancements, organizations can continue to optimize their operations while continuing to increase the scope and revenue potential of their businesses ([Figure 4.7](#)).



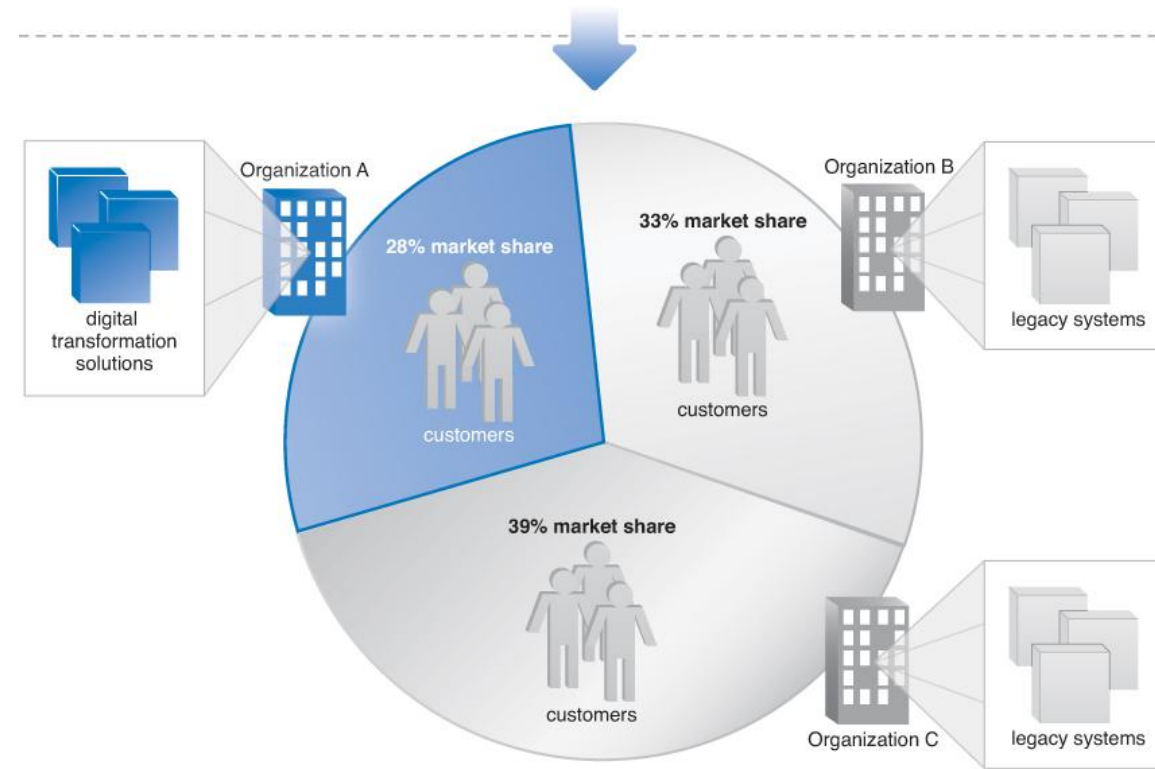
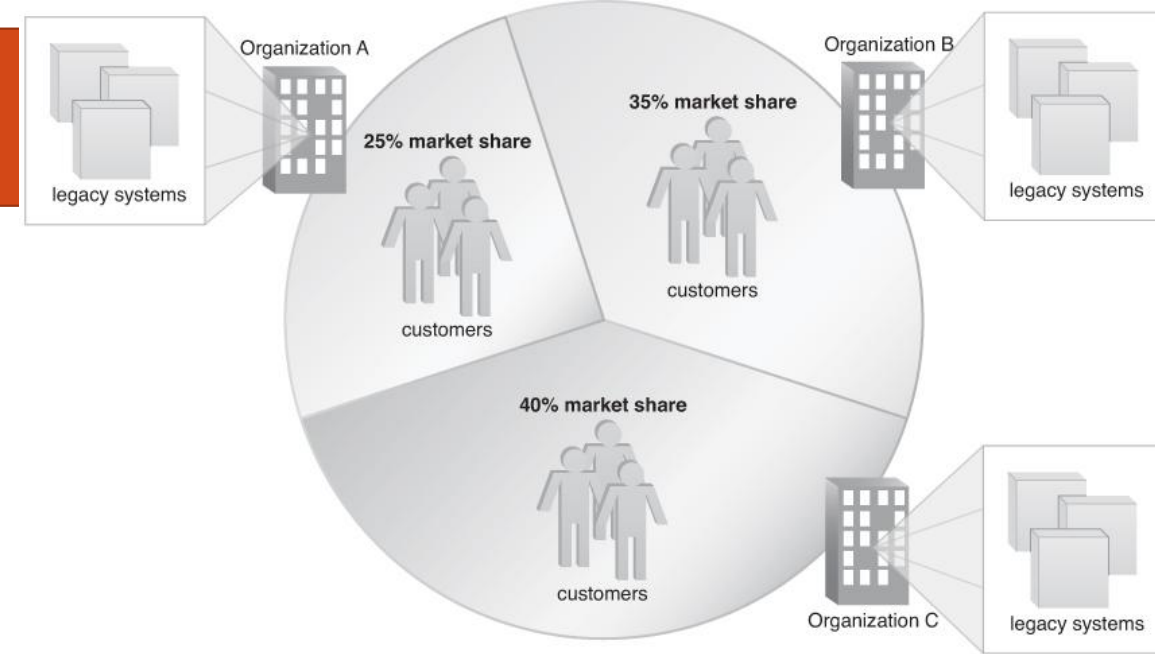


Figure 4.7

Organization A aims to increase its market share by 3% as a result of a successful digital transformation whereby it plans to improve existing services and products and introduce new services and products into the marketplace.

Course: Digital Transformation

Part II

Chapter 5. Common Risks and Challenges (What Are the Pitfalls?)

Pr Habib Fathallah, PhD ECE



[A Field Guide to Digital Transformation, Roger Stoffers, Thomas Erl.](#)
[Published by Addison-Wesley Professional, Digital Transformation](#)

Pr Habib Fathallah, University of Carthage, Tunisia, TextBook: RogerThomas

Chapter 5. Common Risks and Challenges (What Are the Pitfalls?)

- Poor Data Quality and Data Bias
- Increased Quantity of Vulnerable Digital Data
- Resistance to Digital Culture
- Risk of Over-Automation
- Difficult to Govern





Due to the potential scope of a digital transformation, the risks and challenges associated with this type of effort are magnified and need to be carefully assessed to avoid transforming an organization in the wrong direction.

This chapter covers the following common risks and challenges associated with carrying out digital transformation:

- ☐ Poor Data Quality and Data Bias
- ☐ Increased Quantity of Vulnerable Digital Data
- ☐ Resistance to Digital Culture
- ☐ Risk of Over-Automation
- ☐ Difficult to Govern

Poor Data Quality and Data Bias

As previously explained, digital transformation solutions are heavily data-driven.

The data intelligence utilized and produced by these solutions ends up influencing or even determining the outcome of key business decisions.

The quality and success of the decision-making is directly proportional to the quality of the data intelligence upon which the decisions are based.

A constant, overarching concern associated with digital transformation is the completeness, accuracy and orientation of the data used to produce data intelligence.

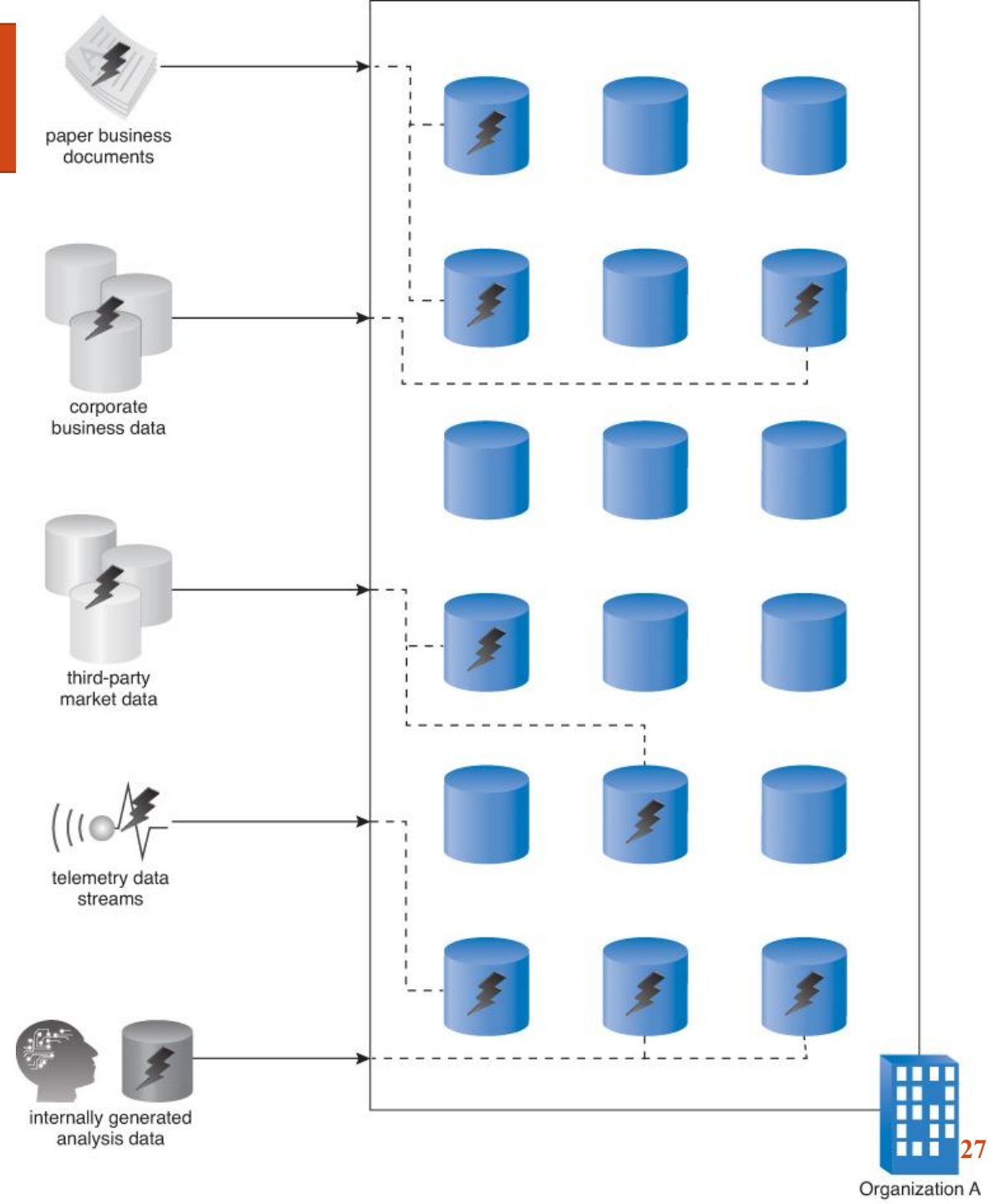
If the quality of the data is poor or sub-par, it can lead to misleading analysis results or missed opportunities to discover new insights ([Figure 5.1](#)).

Another related concern is data bias, whereby some factors within the data are more heavily weighted than others, leading to skewed results and analytical errors.

Figure 5.1

Low-quality data and data bias can creep into almost any type of data collected by an organization.

If undetected, it can make its way into a range of internal corporate repositories where it can influence data intelligence used as input for both manual and automated decision-making.



Increased Quantity of Vulnerable Digital Data

A digital transformation initiative will introduce higher volumes of business data and will often encourage a greater digital presence online.

This can expose significantly greater amounts of an organization's business data to the outside world.

This can, consequently, increase the risks that such data is accessed and manipulated by unauthorized parties ([Figure 5.2](#)).

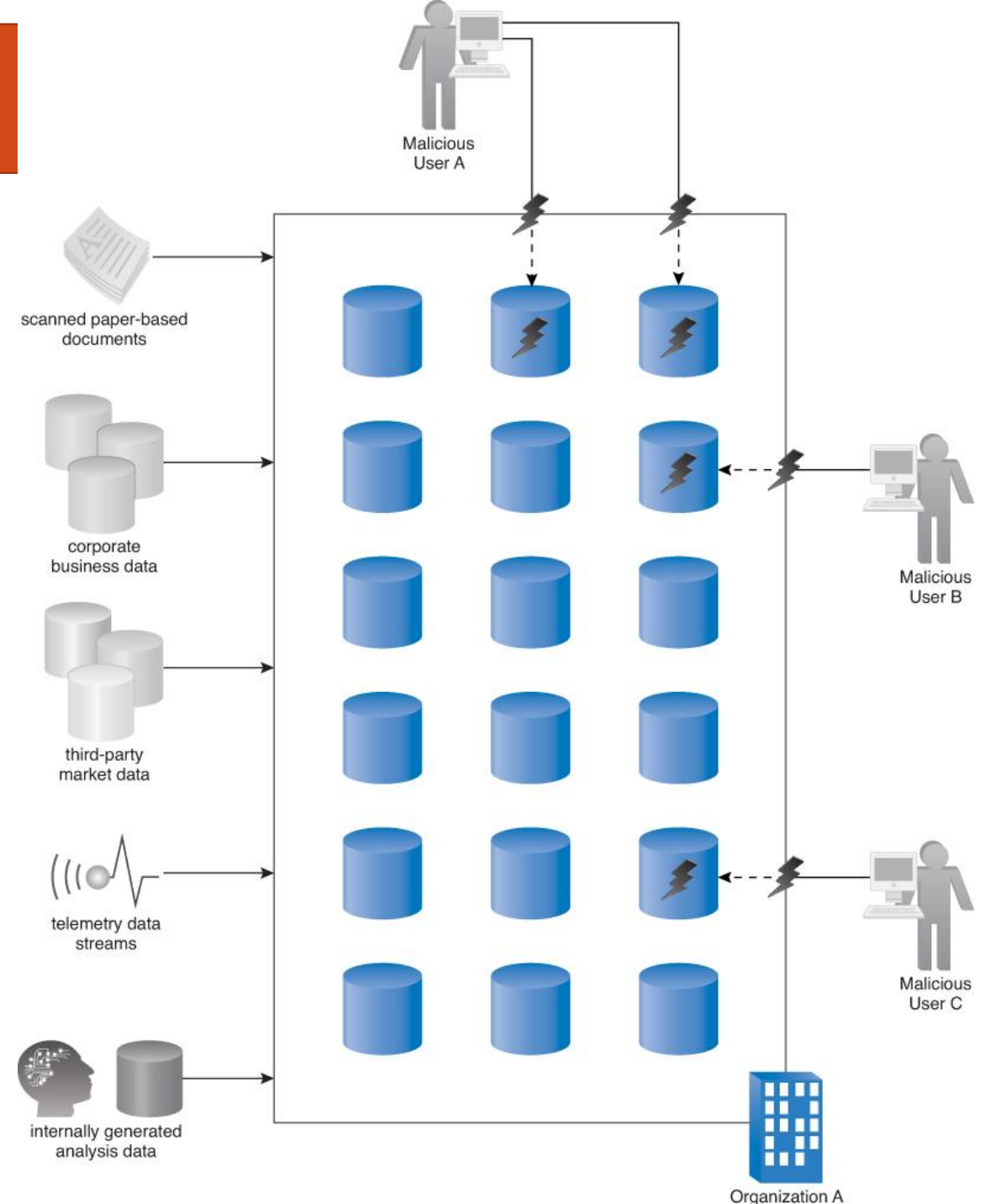
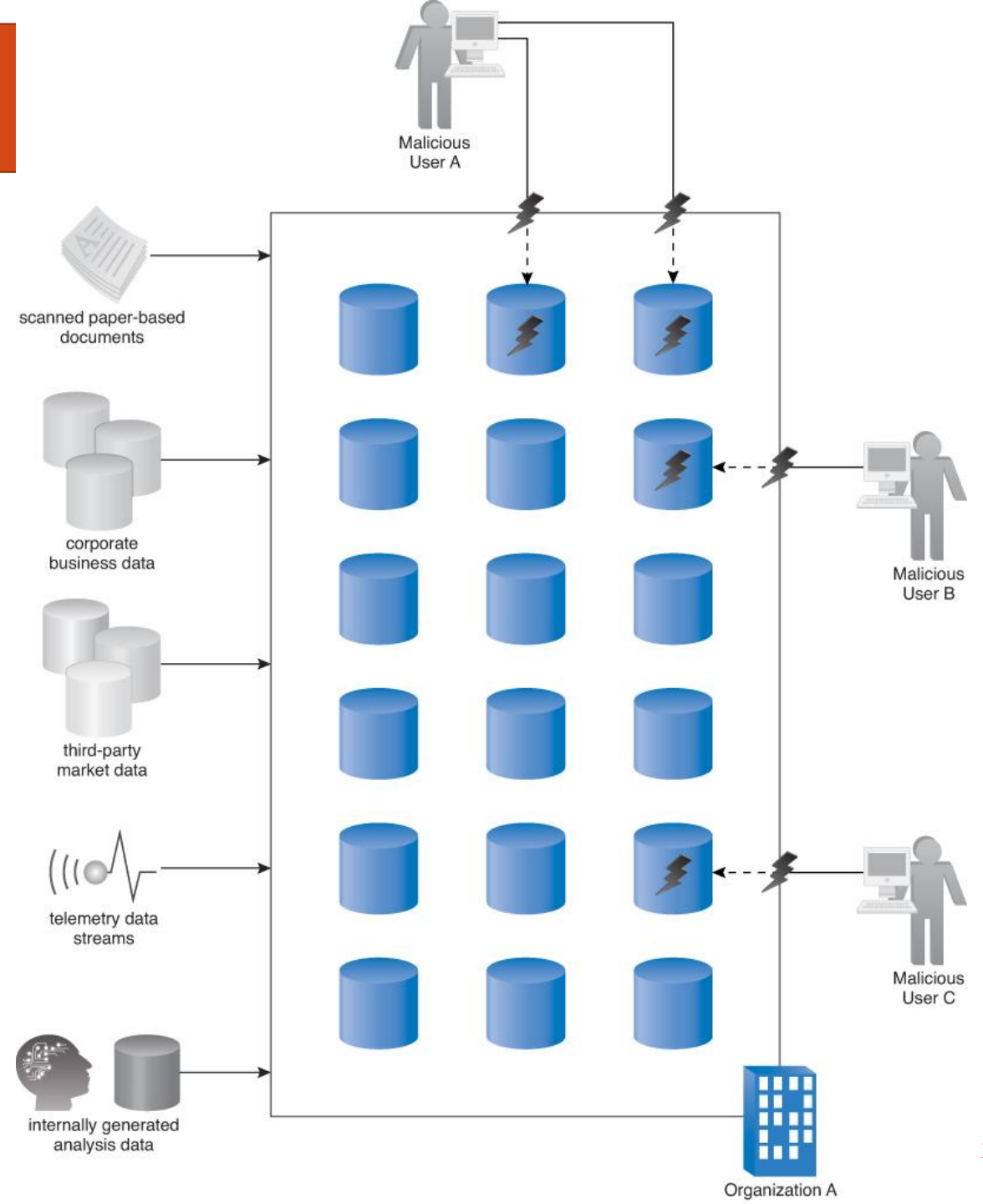


Figure 5.2

The increasing quantity of data flowing into an organization can correspondingly increase the opportunities for malicious users to gain unauthorized access.



Resistance to Digital Culture

A digital transformation aims to change and improve an organization as a whole.

This can include transforming:

- ☐ the nature and scope of an organization's business
- ☐ how humans work and what tasks they are assigned
- ☐ how organizations carry out their day-to-day operations
- ☐ the extent to which automation becomes part of an organization's operations
- ☐ Such an initiative will naturally also transform the overall culture of an organization.

For example:

groups and departments that previously did not need to communicate often may now need to collaborate on a regular basis departments that had full authority over their respective business domains may now lose or need to share that authority employees that previously performed manual tasks may be replaced with new automation technology capable of performing those tasks faster and at a lower cost (Figure 5.3)

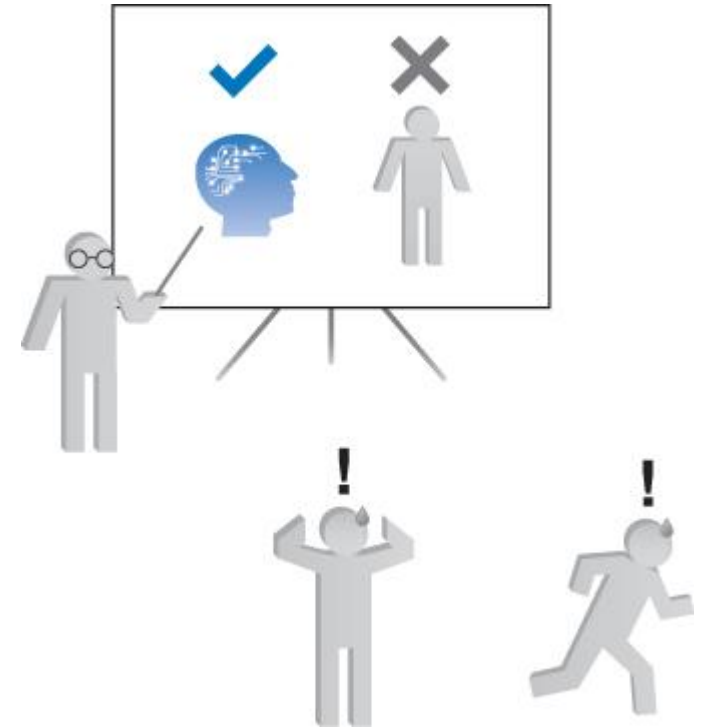
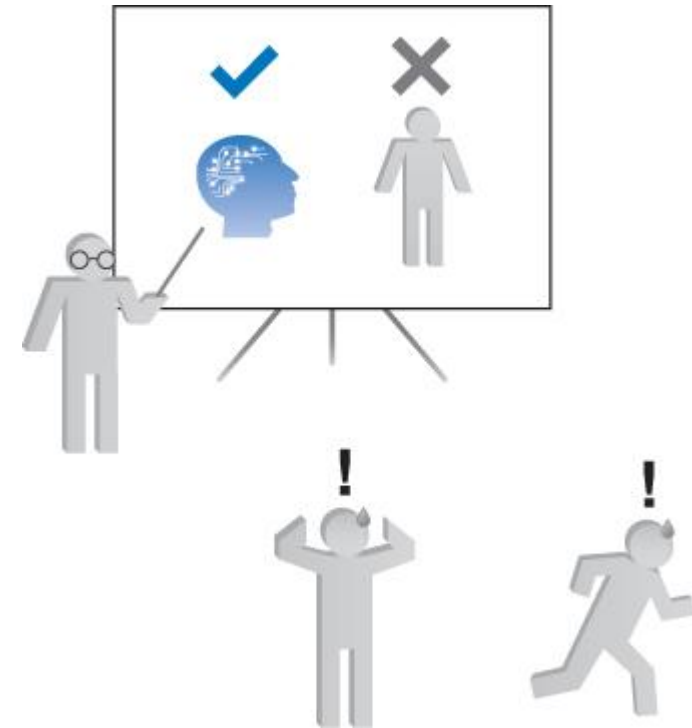


Figure 5.3

Various staff may initially resist digital transformation.

It is common for there to be concerns about job loss and a move toward a culture that can appear to be “colder” and more calculating when it comes to decision-making.

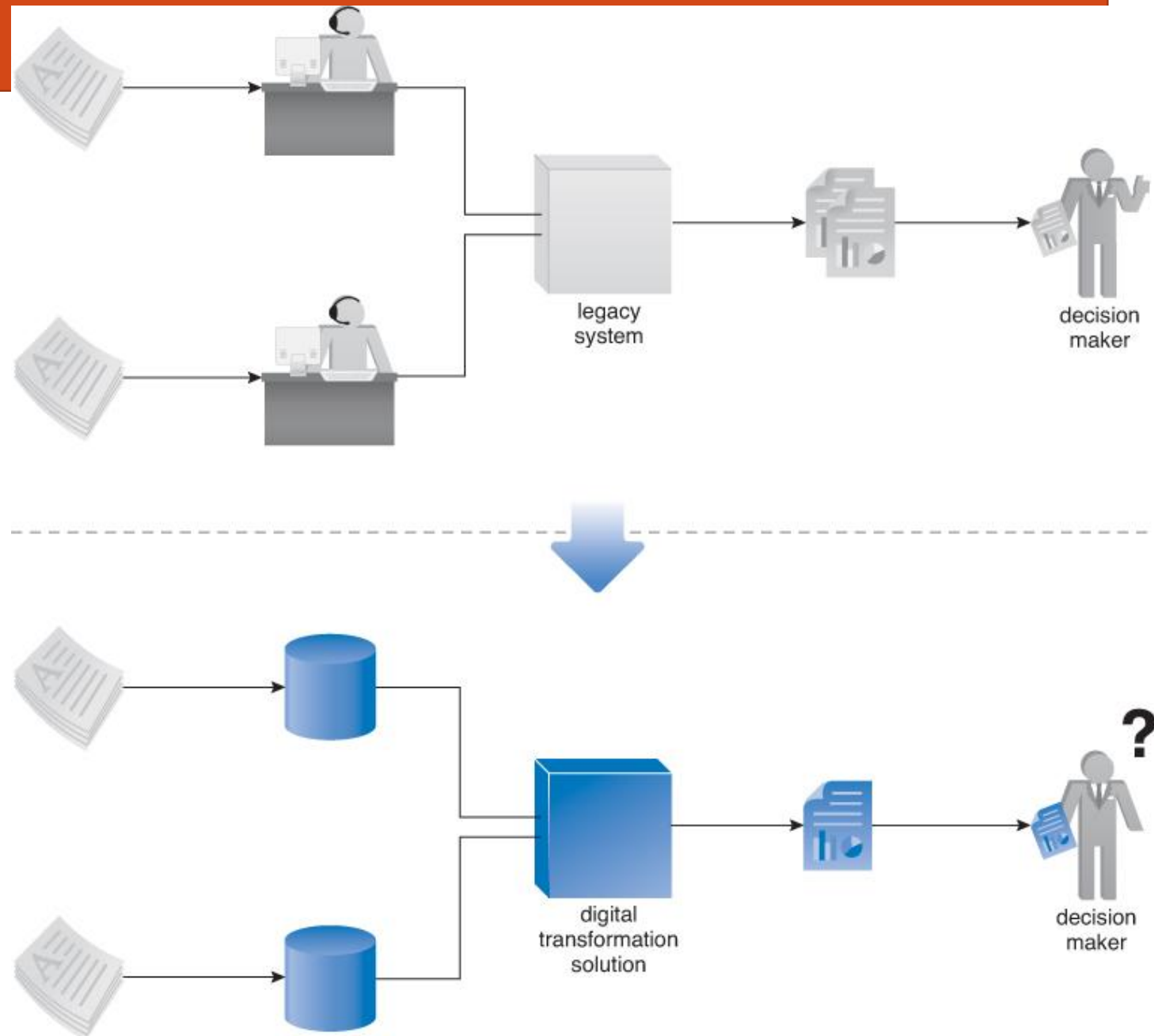


Risk of Over-Automation

The digital transformation market offers a variety of new opportunities to introduce automation into business operations.

In many cases, the benefits are clear and tangible, especially in relation to cost savings and improved time-to-value benefits.

However, there are often reasons as to why some automation options may simply not be suitable, either now or in the future.



Risk of Over-Automation

For example:

- Many of the new automation opportunities are driven by the availability of analytical data that is used to feed into new solution logic.
- The quality or maturity of the data itself may not yet be sufficient to warrant replacing manual tasks with automation ([Figure 5.4](#)).
- This is a case where the automation of certain tasks may need to be delayed until the necessary input data is assessed and considered ready.

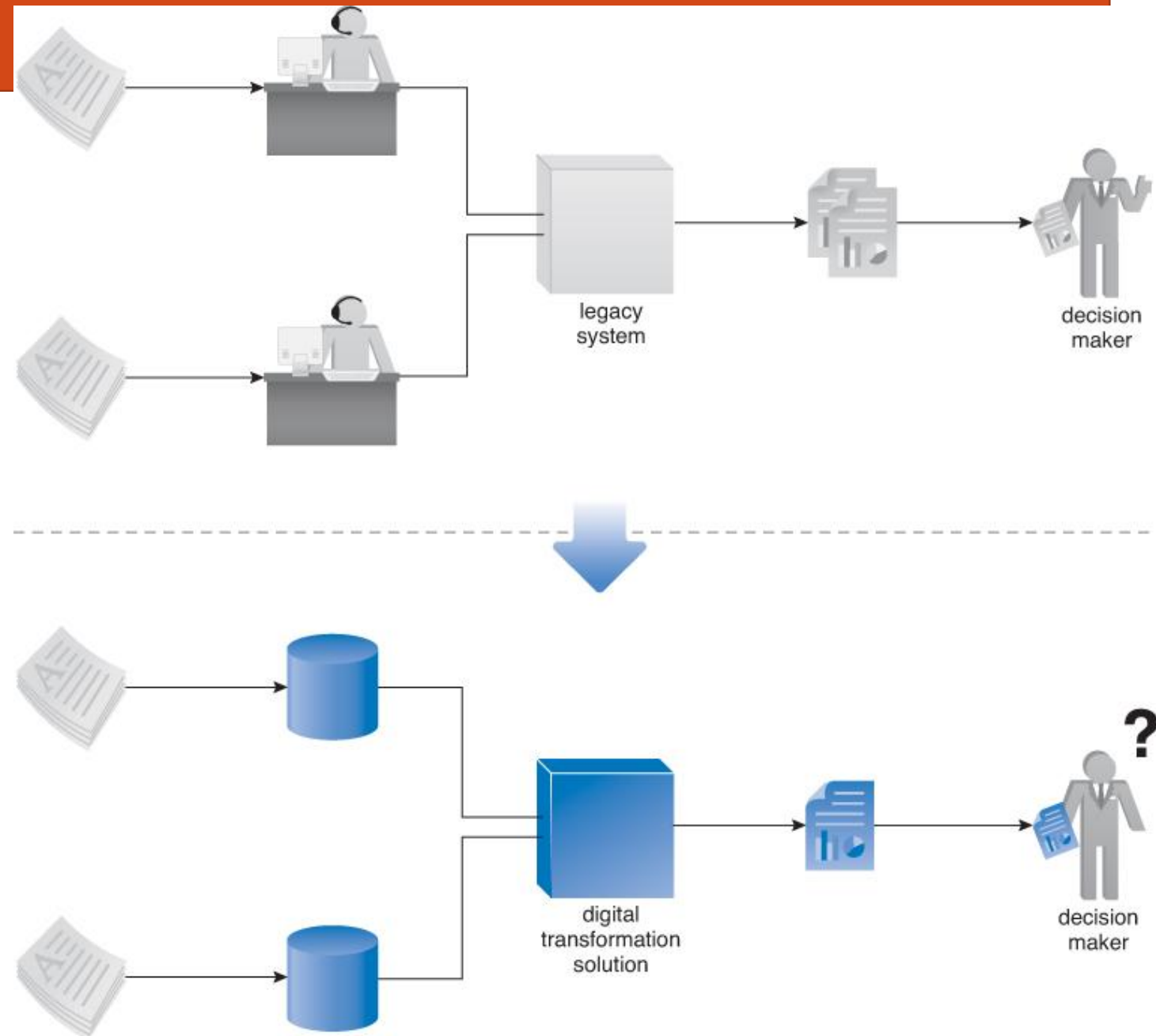
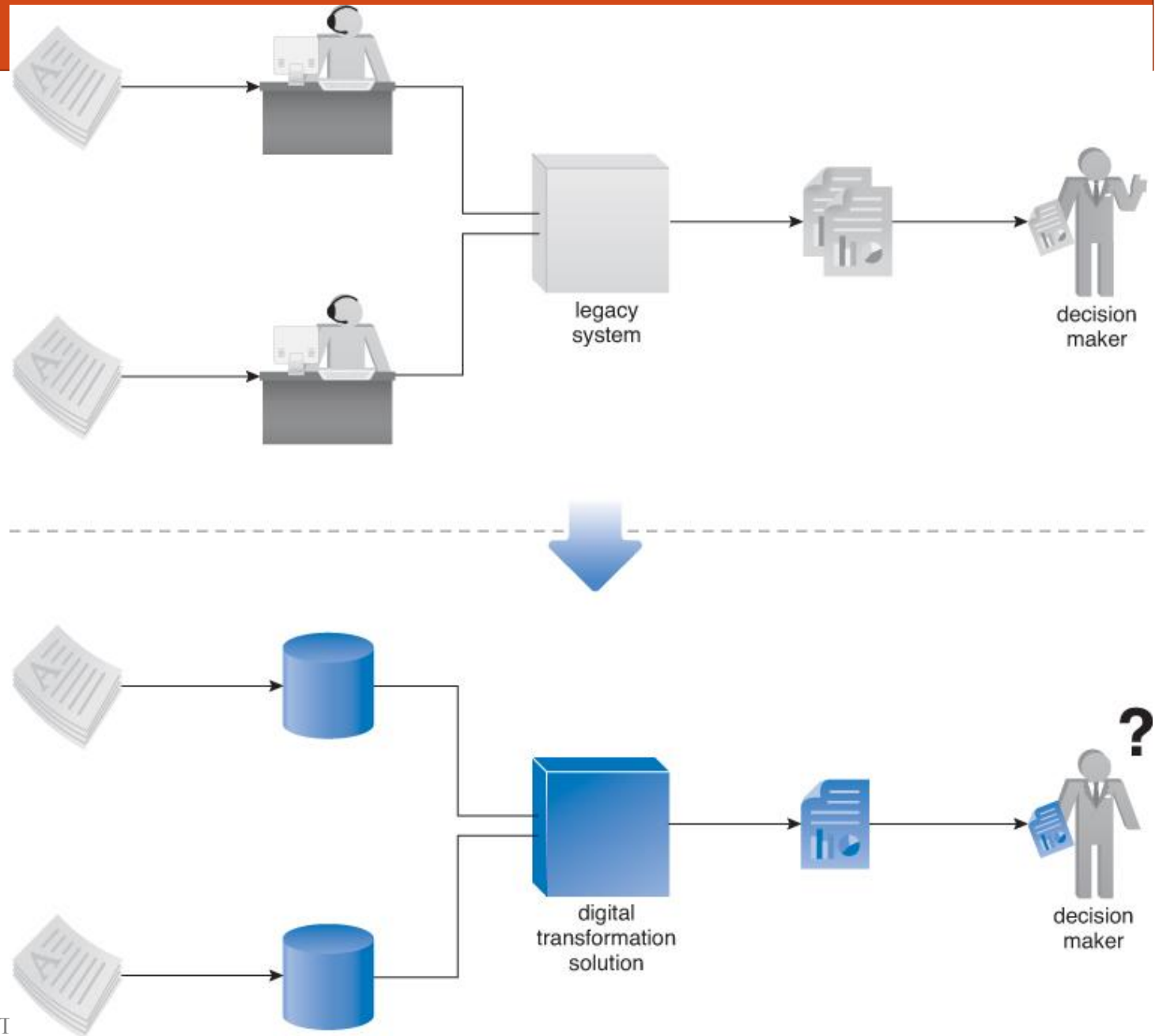


Figure 5.4

A legacy accounting system that previously relied on the manual processing of paper business documents is replaced with a fully automated digital transformation solution.

However, the resulting reporting produced is not valuable because the solution was unable to replicate the quality of the human data entry.

A better solution may have been to still digitize the paper business documents while retaining the involvement of the human resources.



- 
- The organization itself may not have reached a sufficient level of maturity to establish some forms of automation on a broader scale. Perhaps the necessary levels of collaboration have not yet been achieved to create the required level of organizational alignment, or perhaps there is too much resistance to the “digital culture.”
 - Another consideration is the quality of existing business processes. Automating poor business processes does not address their shortcomings and can even reduce the motivation to fix their problems once new automation is in place.

Difficult to Govern

Governance is a key success factor in many IT initiatives to ensure that the introduction of new and improved technologies achieves an organization's tactical and strategic goals.

However, the potential governance scope of a digital transformation effort can be unpredictable and daunting and will go well beyond IT (Figure 5.5).

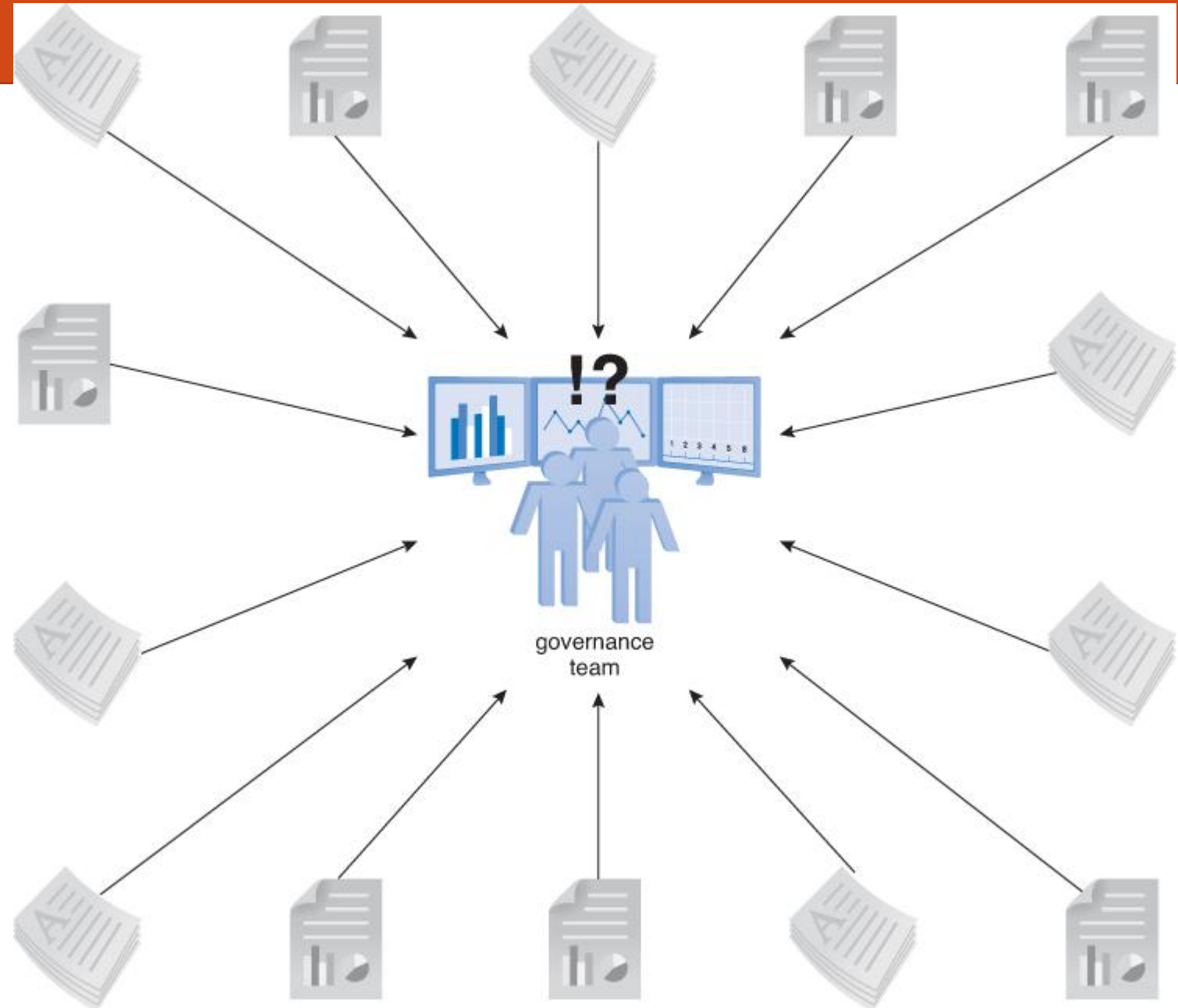
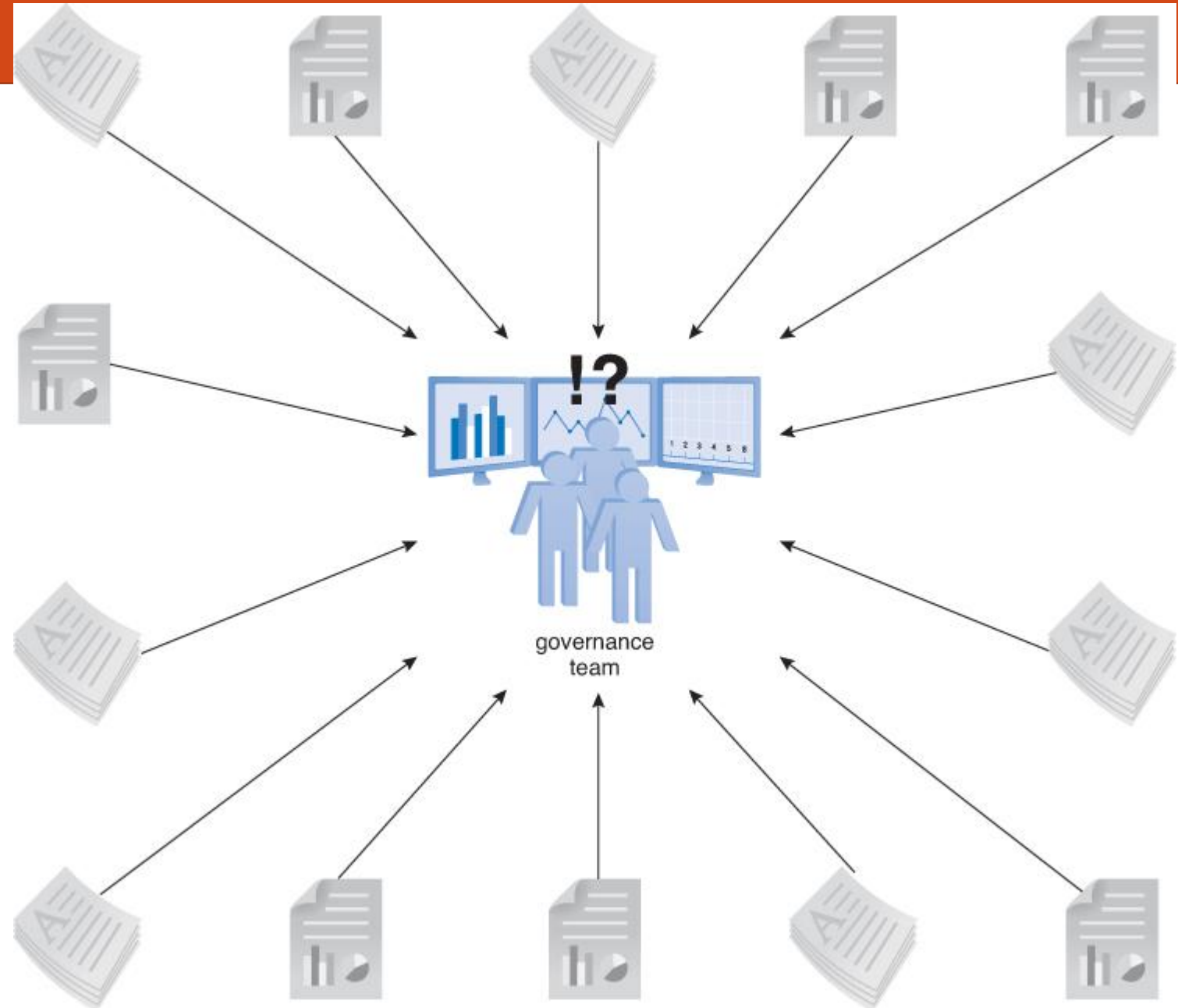


Figure 5.5

A governance group comprised of business and technology professionals is responsible for the governance of an organization's digital transformation.

This team may be challenged to stay on top of the many governance processes and precepts that may be established, created and coordinated.





Digital transformation can introduce several far-reaching impacts to both business and IT that need to be tracked, assessed and regulated in order for the initiative to proceed and evolve successfully over time.

For example:

- ❑ The merging of existing business processes and models and the introduction of new products and services (that further introduce new business processes and models) need to be carefully orchestrated and evolved to ensure they fulfill expectations.
- ❑ The introduction of new technology platforms can supplant legacy systems entirely, or require new forms of integration with legacy environments.
- ❑ The introduction of data-driven technology solutions can change the design and technology architecture of existing automation solutions that perform common automated tasks.
- ❑ The automation of tasks that were previously performed manually can increase the overall scope and responsibility of IT departments.
- ❑ The new forms of cross-silo collaborations that may be required can lead to different ways of budgeting and staffing projects, as well as new assessment criteria and processes.



A governance team comprised of both business and IT professionals needs to be given the responsibility of overseeing and regulating a digital transformation initiative. This team may require significant resources and training to be fully prepared to govern different life cycle stages of many inter-related projects.

A digital transformation governance framework will often introduce precepts and processes that formalize how different parts of the initiative need to be carried out and how the digital transformation environment itself needs to be maintained.

Chapter 6. Realizing Customer-Centricity

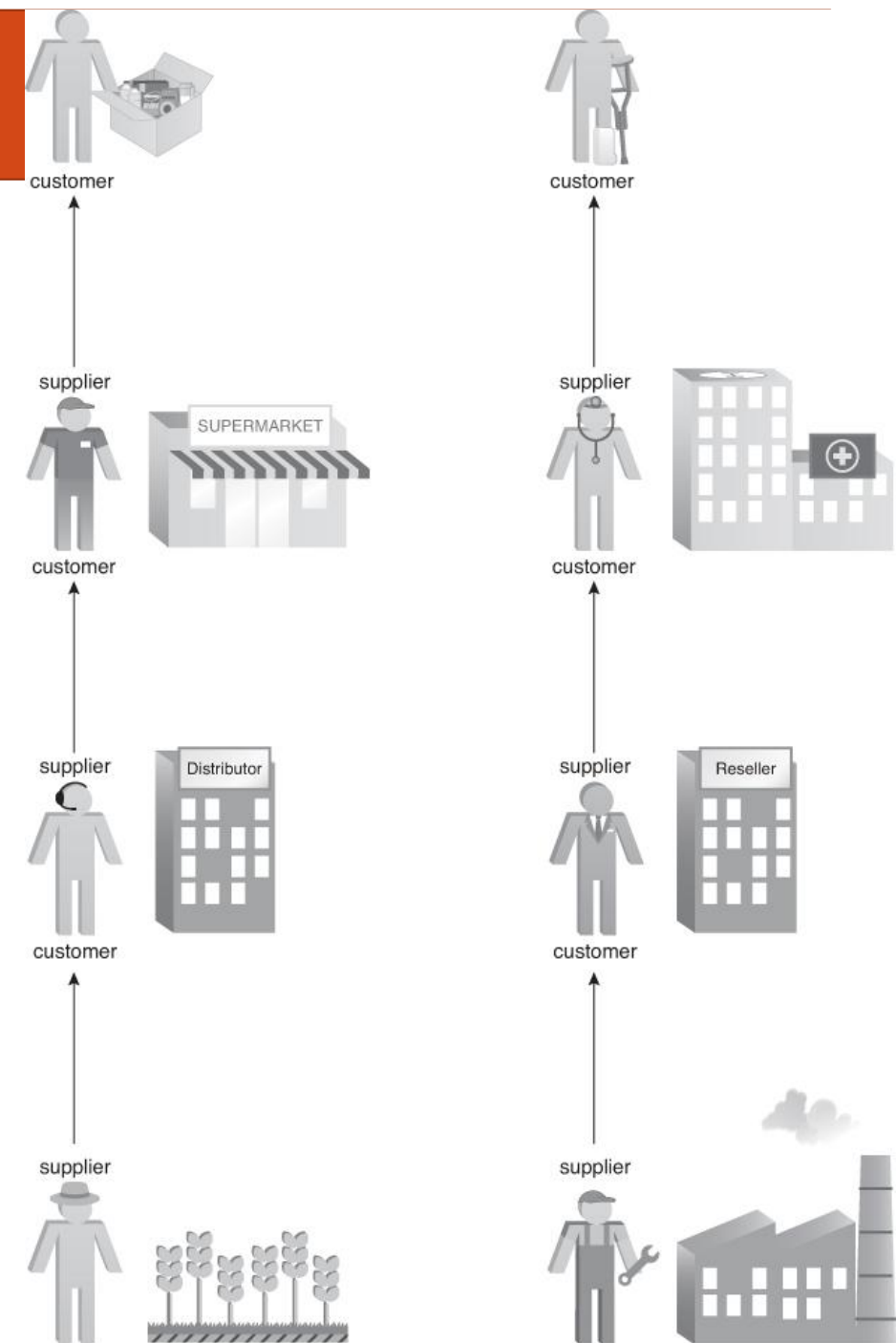


- What Is a Product?
- What Is a Customer?
- Product-Centric vs. Customer-Centric Relationships
- Transaction-Value vs. Relationship-Value Actions
- Customer-Facing vs. Customer-Oriented Actions
- Relationship Value and Warmth
- Single vs. Multi vs. Omni-Channel Customer Interactions
- Customer Journeys
- Customer Data Intelligence

What Is a Product? What Is a Customer?

Figure 6.1

Different supply chains involving different parties, several of which assume both customer and supplier roles.



Product-Centric vs. Customer-Centric Relationships

Almost any type of organization is required to interact with customers.

Traditionally, the emphasis of organizations has been primarily on the quality of the product it offers.

The assumption here is that the greater the value (or perceived value) of a product, the greater the satisfaction of the customer acquiring the product.

This is a product-centric organizational model.

In a product-centric organization the product is the primary business entity around which many parts of the organization are built and structured.

When an organization has several products, there has been a common tendency to build a separate organizational domain around each product.

This can lead to silo-based organizational structures that become fragmented over time as each silo carries on its own lifecycle and direction.

Product-Centric vs. Customer-Centric Relationships

A customer-centric organization positions the customer as the primary business entity and then positions other parts of the organization (including relevant products and value-add services) with the goal of maximizing successful interactions and transactions with its customers.

To achieve this goal, many of the previously silo-based organizational departments are required to collaborate. The result is a broad organizational transformation that goes well beyond being “extra nice” to customers and extends into how technology is utilized to build customer-centric automation solutions.

A classic example is a product-centric bank that has organizational silos for its current product portfolio, which may include savings accounts, mortgages, insurance and investments.

This product-centric business model leads to customers having to interact individually with the bank when inquiring about or acquiring each product.

A customer acquiring all four products may, resultantly, end up having to create and manage four separate customer accounts. A subsequent maintenance task, such as updating an address or phone number, may have to be repeated four times.

This approach predictably leads to a sub-optimal customer experience. In this example, customers may have felt as though they purchased four products from four different banks.

This approach is also not ideal for the organization, as it ends up with redundant and out-of-synch customer data and often an inability to gain genuine data analysis insights into customer trends and statistics.



When fostering relationship-centric customer engagements, the customer experience does not end after the purchase of a new product.

Instead, that is often the starting point of what the organization hopes is to become a genuine relationship with the customer and a relationship that is not limited to the customer's interest in a given product (Figure 6.2).

In fact, an organization may even pursue a long-term relationship with a customer who has not even purchased a product, but who perhaps fits the profile of a customer type that is viewed as a potential corporate asset.

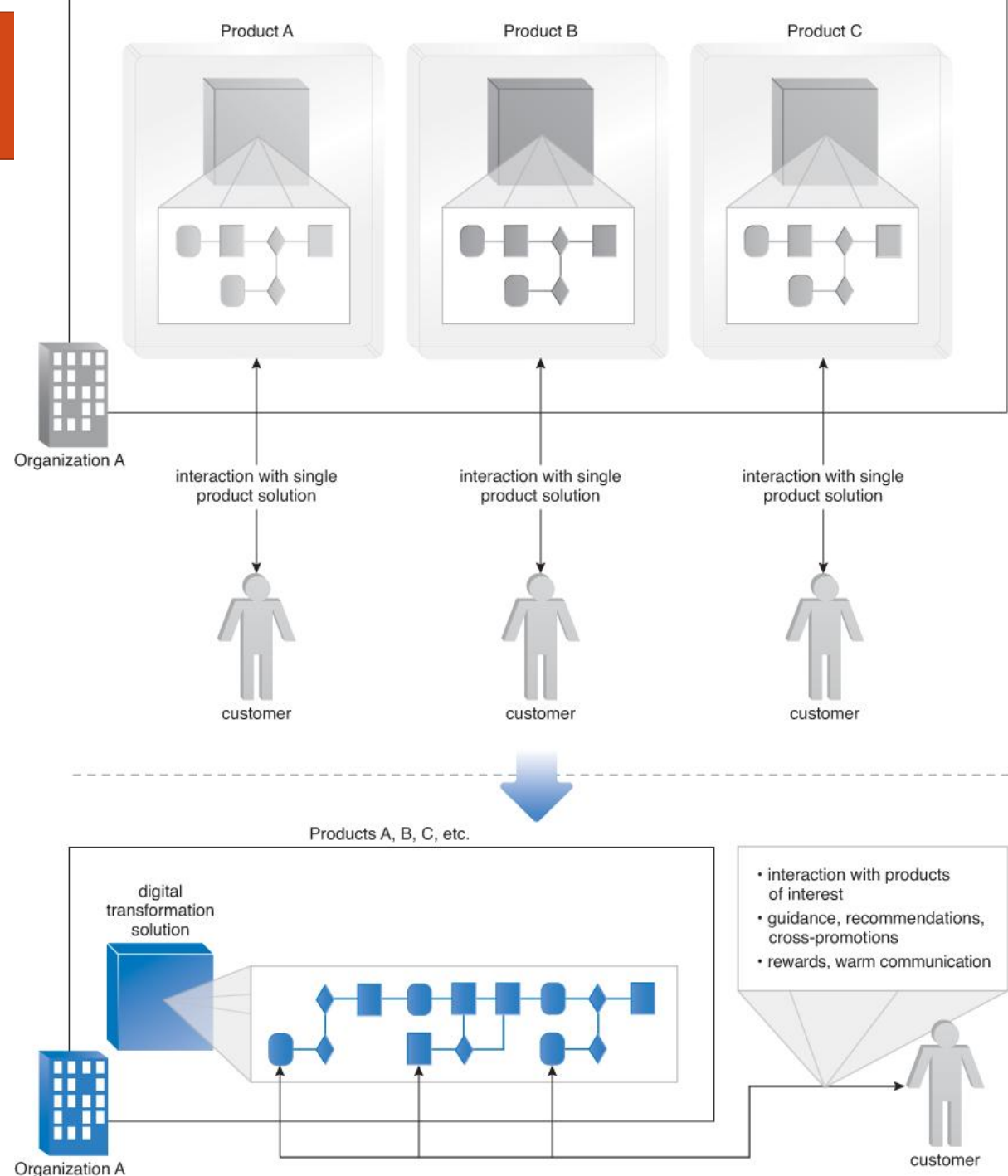
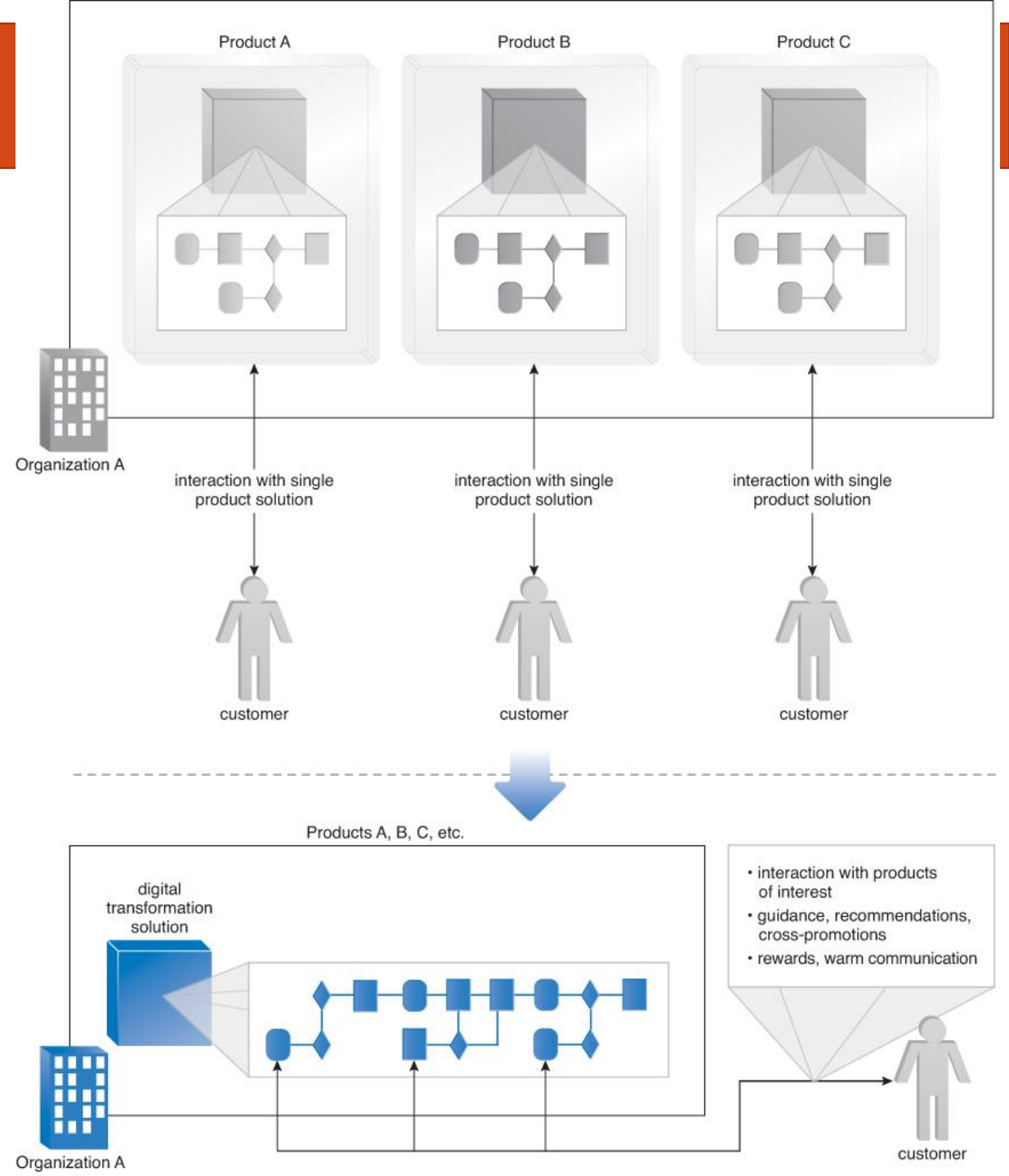


Figure 6.2

A product-centric organization (top) has organizational silos that have established a dedicated business process for each product.

As a result, the customer must interact with each product solution individually.

After the organization becomes customer-centric (bottom), the silos are removed and the business processes consolidated to provide a single interaction experience during which the customer can learn about and purchase any products of interest, and during which the customer is motivated to return on a regular basis.

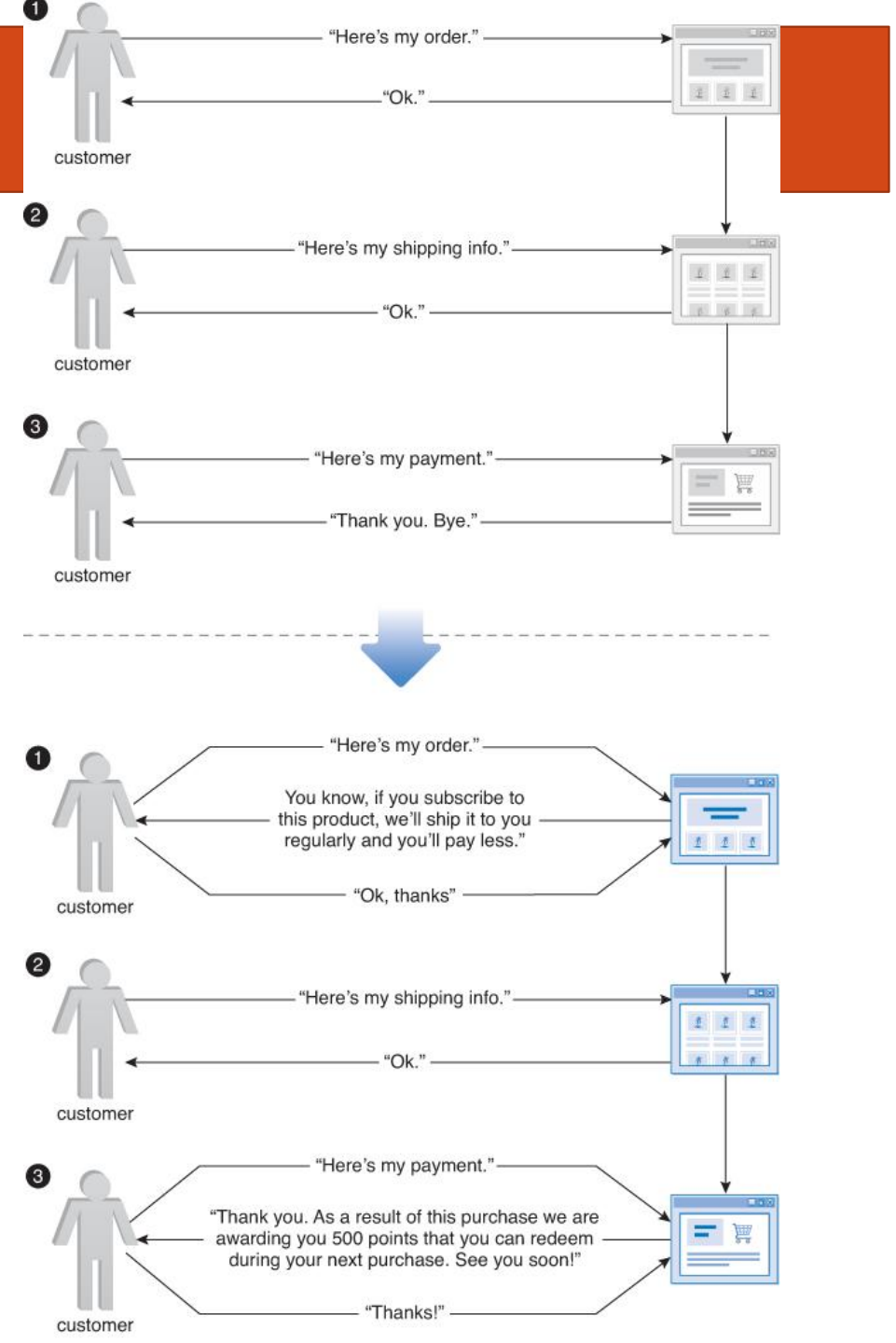


Transaction-Value vs. Relationship-Value Actions

During a transformation toward a customer-centric business model, the individual interactions that comprise business processes, workflows and interaction scenarios are often carefully studied and classified as follows:

- *Transaction-Value Action* – A short-term result from an interaction with a “one-off” activity that is not primarily geared toward making the customer return classifies the interaction as a transaction-value action. For example, the payment step in a product purchase scenario can be considered a transaction-value step that is essentially a one-off action that does not lead to further activities from the customer to increase their value to the organization.
- *Relationship-Value Action* – A long-term result from an interaction that is intentionally geared toward making the customer return classifies the interaction as being a relationship-value action. For example, the aforementioned payment step could be remodeled as a relationship-value activity by awarding the customer with credit or points that the customer can apply toward a future purchase. Each purchase motivates the customer to return again, thereby fostering a relationship.

Note also that not all customer interactions need to be relationship-value based. There may be several steps and actions within a given scenario that do not warrant or cannot support a transformation toward relationship-value. What is of primary importance is that the overall business process is sufficiently improved to have shifted from being product-centric to customer-centric in support of the organization's digital transformation goals ([Figure 6.3](#)).



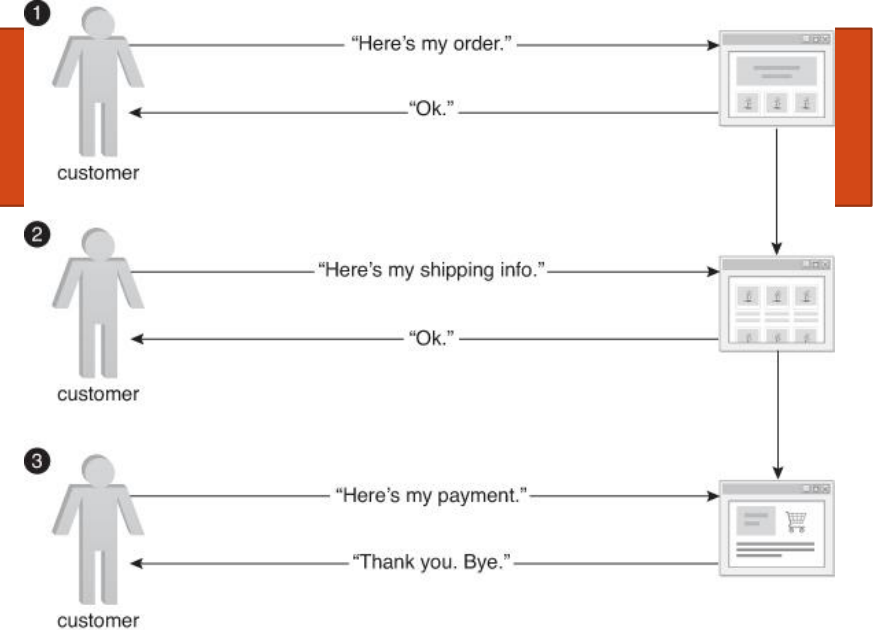
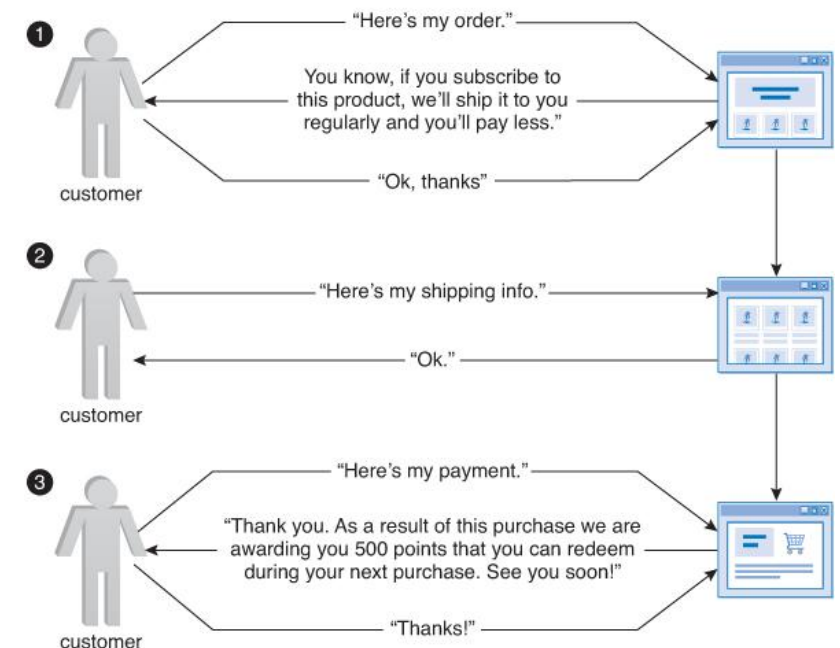


Figure 6.3

An online order business process with transaction-value steps simply receives information from the customer and concludes the transaction (top).

The same online order business process is remodeled so that Steps 1 and 3 become relationship-value actions aimed to increase customer satisfaction and that encourage the customer to return (bottom)



Customer-Facing vs. Customer-Oriented Actions

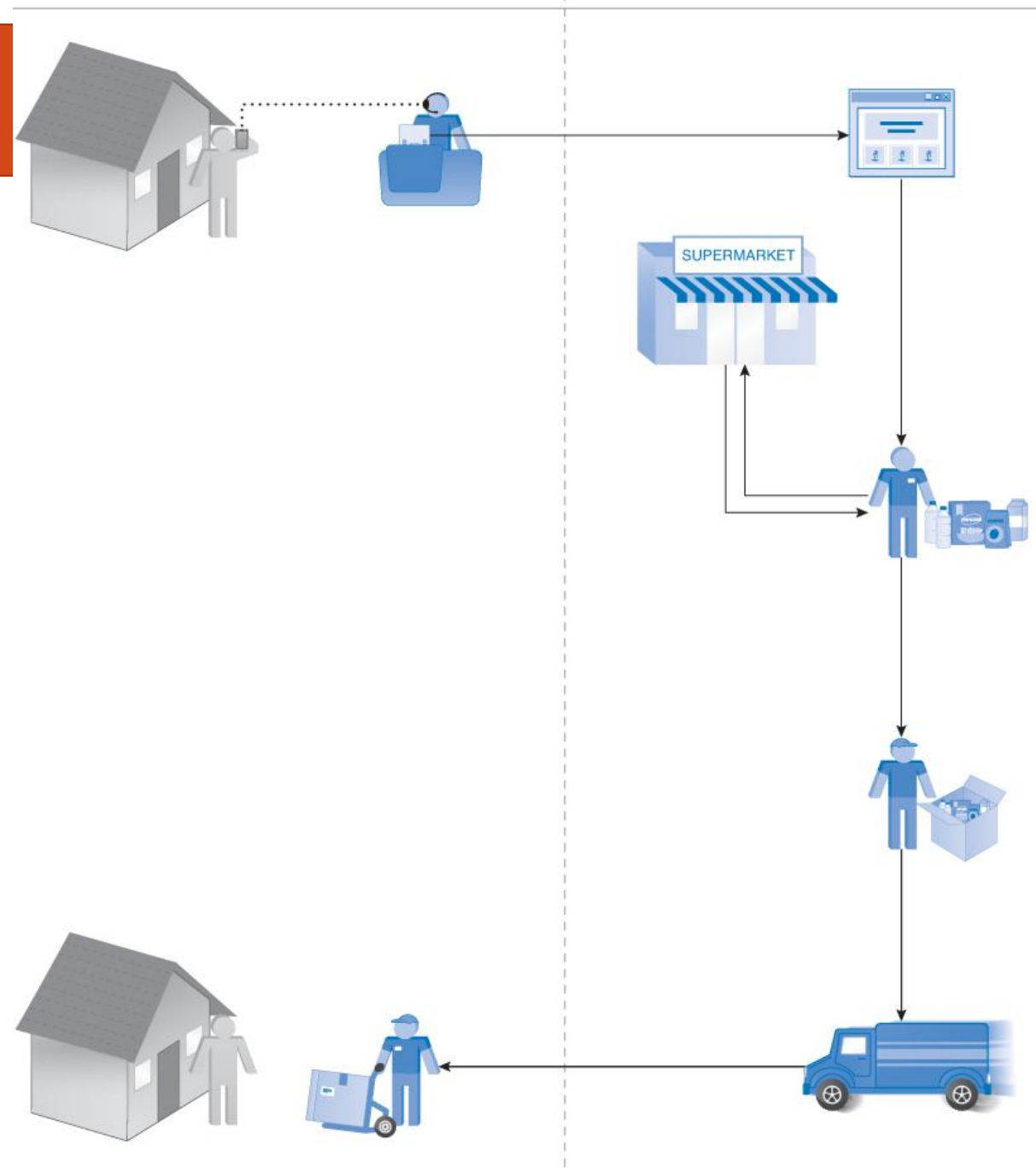
When breaking down business processes associated with customer activity, there is a need to separate actions into two further categories:

- *Customer-Facing Actions* – An action or activity initiated by or resulting from direct customer interaction is a customer-facing action. An example is when a customer interacts with an online store to place an order.
- *Customer-Oriented Actions* – An action or activity associated with assisting a customer or supporting a customer relationship but that does not involve direct interaction with the customer is considered a customer-oriented action.
- An example is when a customer orders a product that is not in stock, and behind-the-scenes steps are taken to process a backorder for the customer.
- Another example is ensuring that solutions used by customers remain consistently operational so that the customer does not have to experience unexpected outages.

When migrating to a customer-centric business model, an effort is made to transition transaction-value actions that are both customer-facing and customer-oriented to becoming relationship-value actions, wherever applicable ([Figure 6.4](#)).

Figure 6.4

When a customer places an order for grocery items to be home-delivered, a series of steps is triggered, only the first and last of which are actually customer facing.



Warmth in Communication

The tone and language of communication used with customers is often, by default, formal, polite and concise. This is because customer communication has traditionally been geared toward professionalism and efficiency, rather than relationship-building.

When determining the *communicative warmth* of a digital transformation solution, a number of factors can come into play:

- The tone of communication may be set or altered, which can affect the language and illustration used in both relationship-value steps as well as transaction-value steps. For example, the customer-facing web pages for an online store may be redesigned to provide statements directed at customers using more casual, friendly or encouraging wording expressed by likeable cartoon characters ([Figure 6.5](#)).

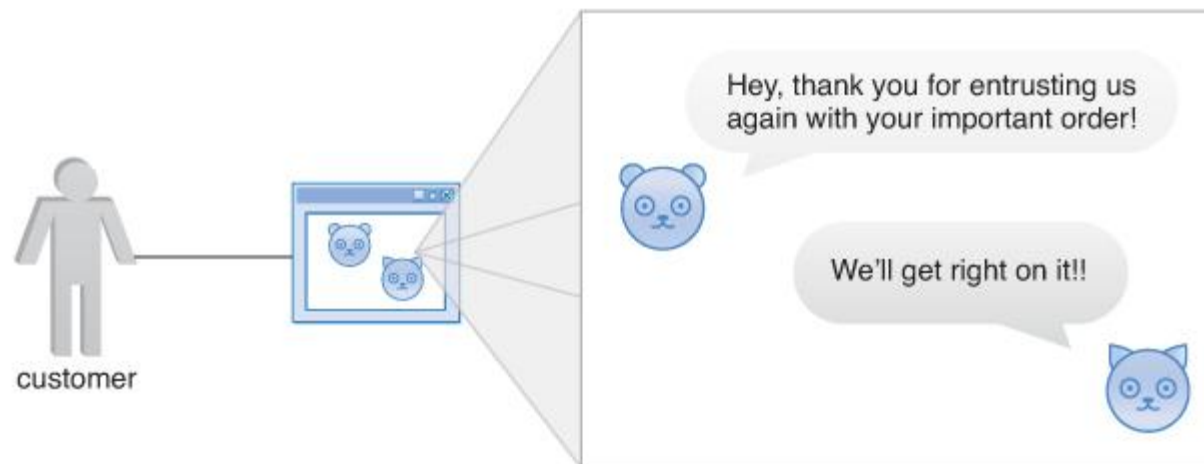
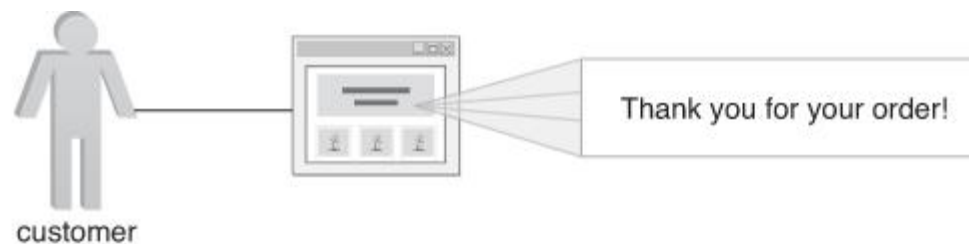


Figure 6.5

An online application is updated to change the tone and language of communication, further supplemented with warm graphics.

- 
- Human phone and chat operators may be retrained to interact with customers in a more friendly manner. This may require learning new scripts as well as developing some new social skills to strike up casual conversation during waiting times.
 - Other customer-facing employees, such as customer service and delivery personnel, may be trained to improve their social skills so as to treat customers with an increased sense of courtesy and genuine interest.
 - The branding of an organization itself may be augmented to appear less cold and corporate. Instead, the branding may be designed to be more personable and relatable to the target customer-base.

Introducing these types of “warmth” alone will likely not turn a given transaction-value action into a relationship-value action. However, applying a level of warmth, where appropriate, does increase the overall customer-centricity of a solution.

Warmth in Proactive Accommodation

There are sometimes opportunities to introduce *proactive warmth* in customer-facing and customer-oriented actions. Sometimes, these actions are carried out in support of a customer's well-being. Other times, these types of actions are carried out due to a necessity to maintain a guarantee to a customer when unforeseen problems arise. Either way, this type of warmth often requires that the solution or the human worker be creative in how they are proactive.

For example:

- A customer places an order from an online vendor and is promised a 3-day shipping period. However, there is an unavoidable one-day delay in retrieving the product ordered and it cannot be provided to the courier in time for the 3-day shipping service to deliver it at the originally promised delivery date. Instead of notifying the customer that the shipment will be late, the solution or a back-end worker proactively decides to use a 2-day shipping method. Even though this action will end up costing the vendor more, resulting in less or no revenue from the sale, keeping its original shipping guarantee to the customer is considered a priority in order to maintain a long-lasting customer relationship.
- A customer places an order from an online vendor for a quantity of six food items. After accepting the order, the vendor realizes that only four of these items are in stock. Instead of issuing a partial refund and shipping a partial order, the vendor replaces the two missing food items with two other, comparable food items. The payment is still partially refunded, but the customer is given complimentary product as a means of mitigating the inconvenience of not receiving exactly what they ordered.

In the former example, the warmth is “transparent” in that the vendor did not add any noticeable warmth to its customer interaction, but instead proactively avoided a situation that could have led to the customer being disappointed and losing confidence.

In the latter example, the warmth is evident to the customer. The vendor also attempts to avoid disappointing the customer but does so by “rewarding” the customer with a gift to offset any negative feelings about the shipment.

In both cases, these extra steps taken by the vendor can be classified as relationship-value actions, as they incur extra effort and expense in support of improving customer- centricity and fostering long-term customer relationships. Therefore, some organizations may utilize their available “customer data intelligence” to analyze and assess customers on a case-by-case basis to determine whether this extra effort and expense are warranted ([Figure 6.6](#)). This type of customer profiling is further explored in the upcoming *Customer Data Intelligence* section.

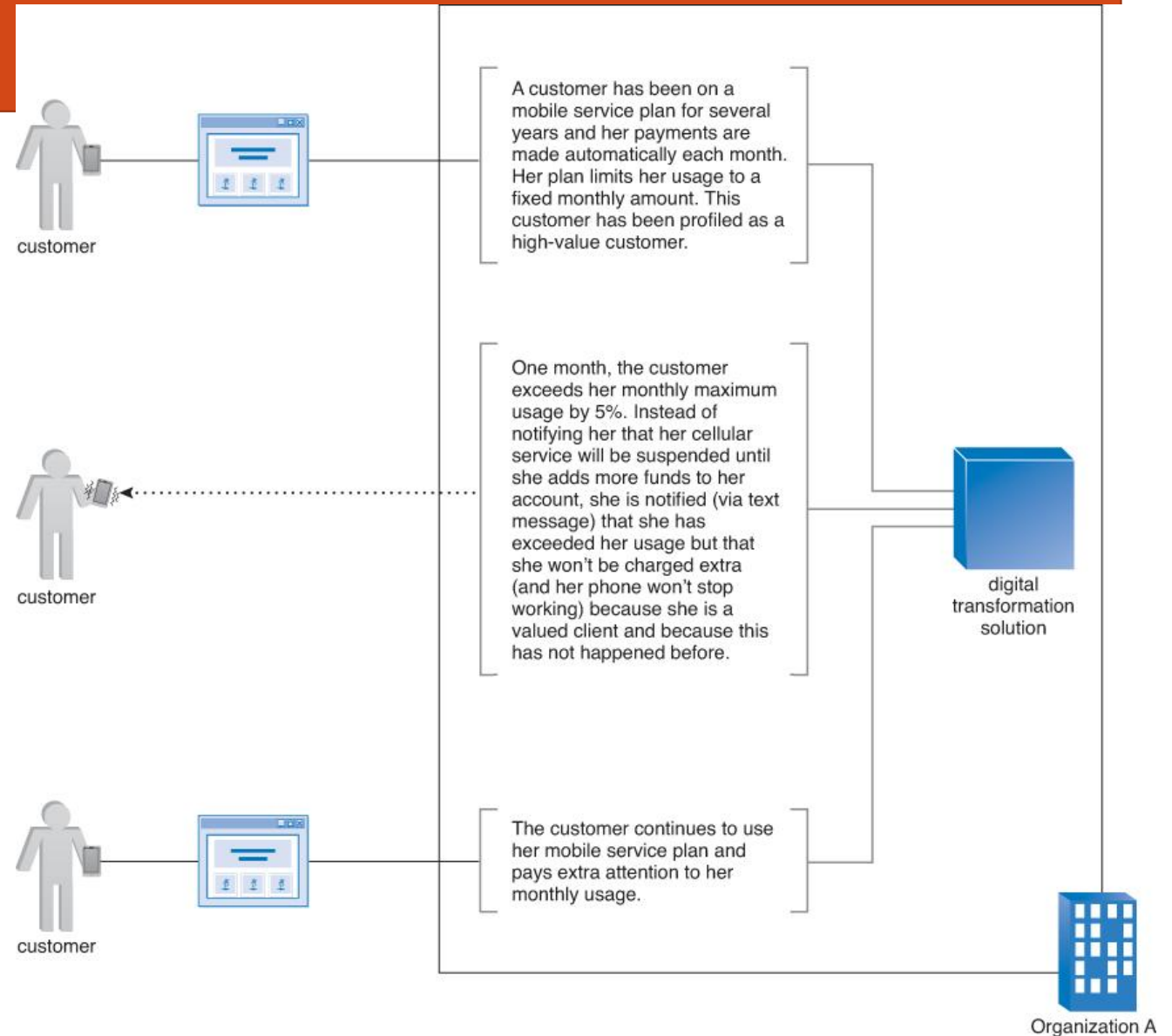
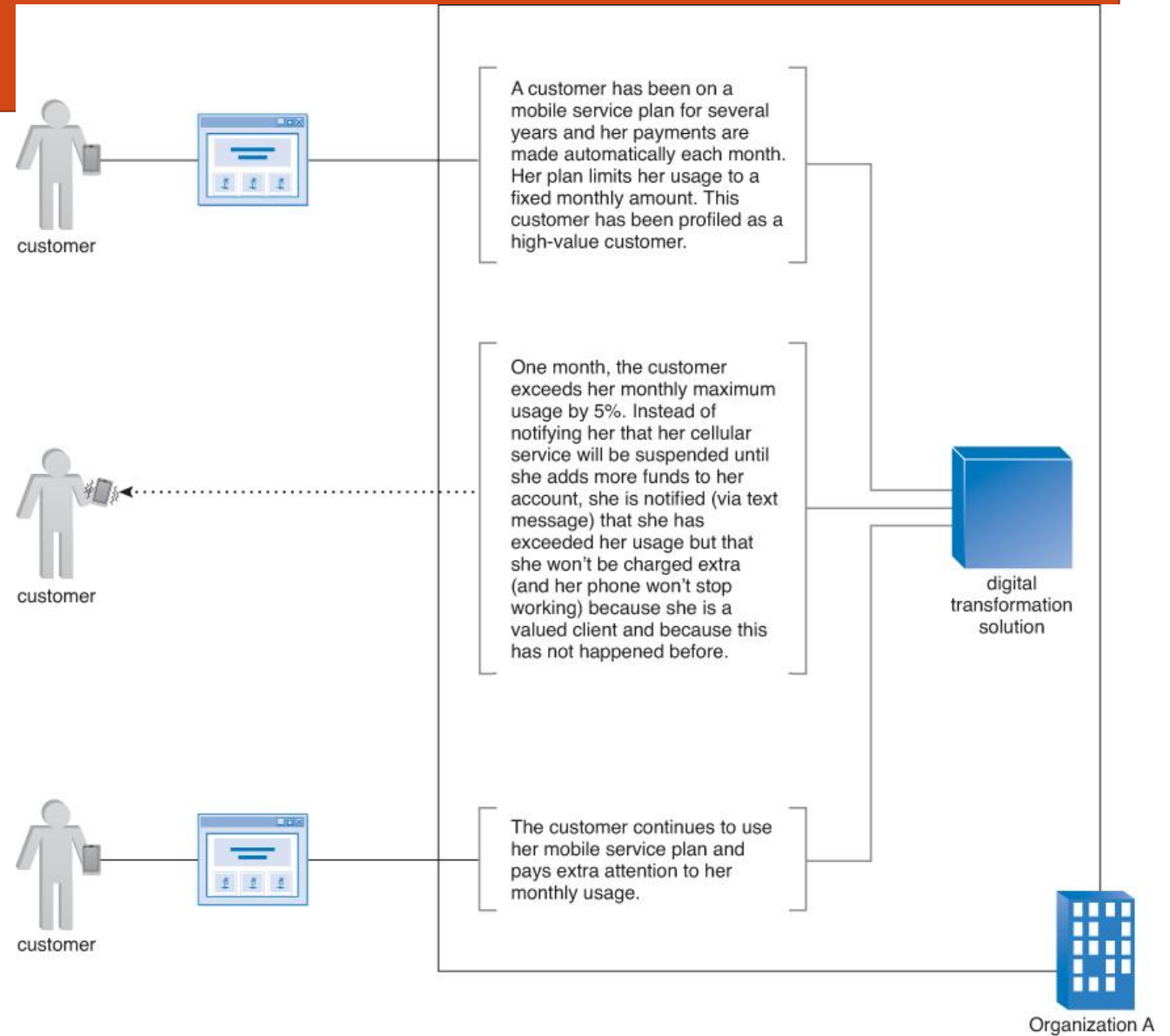


Figure 6.6

A mobile phone company incurs the extra expense of a customer's excess phone usage so as to maintain a positive relationship with what has been profiled as a valuable customer.



Warmth in Customer Rewards

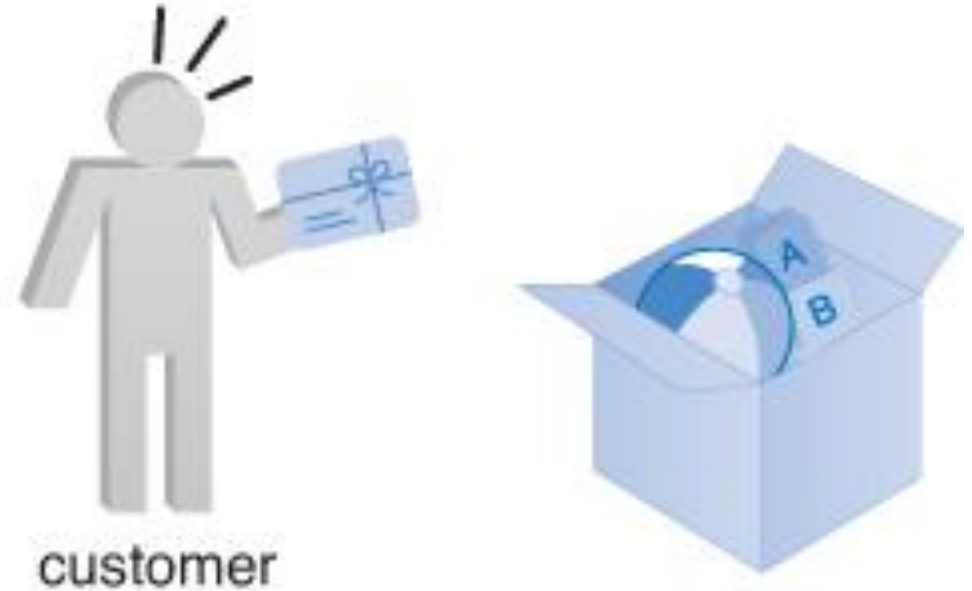
Organizations can increase the *rewardful warmth* of customer interactions by awarding gifts and other types of rewards in recognition of customer value or loyalty. A classic example of this is the reward points system often used with online stores, whereby the customer receives points subsequent to each purchase that can be later redeemed for monetary value. This simple and proven system is effective in encouraging repeat business, but can also become a predictable part of transactions, to the extent that it is not genuinely “warm.”

Often, the most effective types of rewards are those that are unexpected by the customer. For example, a mobile service company applying a bonus credit to the last bill of the year in recognition of a customer’s one-year anniversary. Or, a law firm sending a food basket to a valuable client after successful negotiations. Unexpected and generous gestures such as these are considered relationship-value actions, as they can provoke positive emotions in customers that will be remembered ([Figure 6.7](#)).

Figure 6.7

A customer is surprised to discover a valuable gift certificate in a package with the items originally ordered.

"I wasn't expecting this!!"



Single vs. Multi vs. Omni-Channel Customer Interactions

Customers are generally able to complete the same task by interacting with an organization in different ways. For example, traditionally, a customer would be able to place an order by calling a store or by visiting the store in person. Each interaction option is referred to as a *channel*.

If the store would not take orders over the phone and would only offer transactions for those visiting the store in person, its ordering business process would be considered *single-channel*, because only one interaction option is offered to the customer. In the aforementioned case where the store also offered phone-in orders, its ordering business process would be considered *multi-channel*.

For most customer interaction scenarios, single-channels are rare and multi-channels are common. For example, a customer can purchase an item from a physical store just as easily as visiting the company's online store. The problem with multi-channel interactions is that the customer experience with each channel is usually completely independent. If a customer visits a store and asks to put a product on back-order, a multi-channel environment would not necessarily carry over that back-order to the customer's online store account.

Let's imagine the customer originally intended to pick up the back-ordered item from the store, but later decided to arrange for the item to be shipped to the customer's home. With a multi-channel ordering process, the customer would need to phone the store to cancel the original back-order and then log onto the online store to place a new back-order for the product to be shipped to the home address.

Single vs. Multi vs. Omni-Channel Customer Interactions

Digital transformation advocates an *omni-channel* approach, whereby customer interactions across multiple channels are kept in synch. Digital transformation further advocates a *multiexperience* approach whereby customers have a wide range of access methods made available to them.

Examples of the types of channels commonly supported by *multiexperience* environments include:

- web-based online accounts
- phone and virtual communication
- online chats
- email exchanges
- in-person visits
- paper-based letter or document exchanges

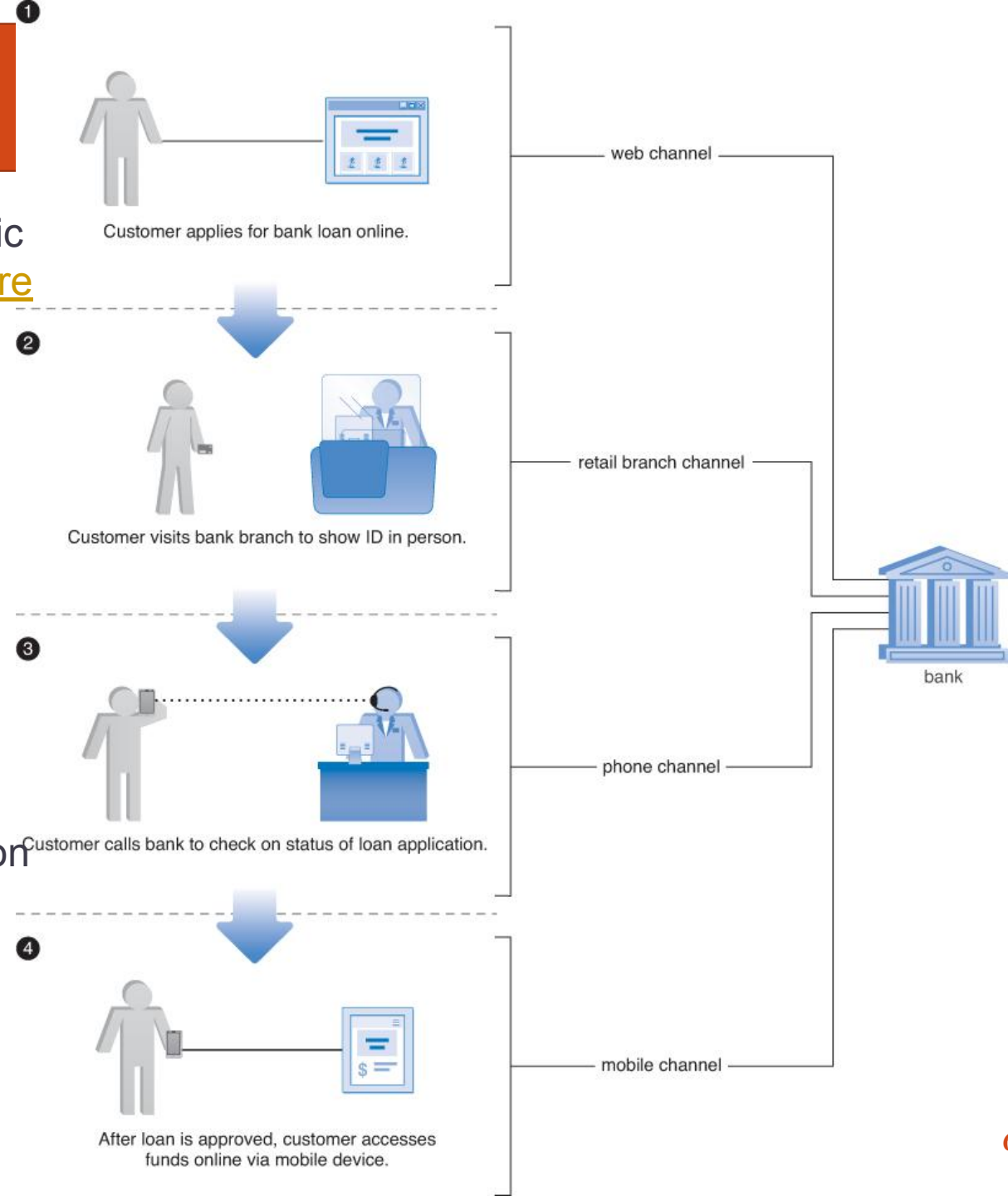
Examples of the types of devices commonly supported by *multiexperience* environments include:

- laptops and workstations
- smartphones and tablets
- smart watches and other wearables
- smart TVs
- voice-activated devices

Omni-channel support is a key aspect of customer-centric solution design that provides the following benefits ([Figure 6.9](#)):

Figure 6.9

A customer interacts with a bank to apply for and eventually collect a loan. Each step involves interaction via a different channel. Because the bank's loan solution is designed to support omni-channel processing, the customer never needs to repeat any part of the interaction



- Customer satisfaction is increased because customers become aware of the fact that they can switch from one channel to another without having to repeat previous actions. This is especially important with customer interaction scenarios that can take longer periods of time, in which case customers especially appreciate the flexibility of being able to discontinue on one channel and then continue later on another.
- The organization's labor and effort associated with the customer interactions is streamlined because there is less need to accommodate changes, cancellations and redundant process steps across multiple interaction scenarios.
- There are increased opportunities to add relationship-value actions to the overall customer experience because some channel mediums can provide more enhanced customer-centric features than others. For example, a customer that initially visits a store and then switches to an online interaction can now be served online ads, recommendations and rewards that may not be possible during the in-person visit.

Customer Journeys

A customer journey is the collection and sequence of steps and actions a customer carries out when interacting with an organization, a product or a service, in order to achieve a goal.

For example:

- A customer that visits an online store may first interact with the store to browse available products. Then, after choosing a product to purchase, the customer interacts with the shopping cart to choose a shipping option, and finally the customer will interact with the store to process the payment for the order. This customer journey was comprised of these interactions with the goal of buying a product. In this case, the customer journey lasted perhaps a few minutes.
- A customer that visits a mobile phone store may first interact with a sales clerk to purchase a phone and may then interact with a phone support employee to choose a phone number and have the phone registered. The customer may further interact with a website to have the phone activated. In this case, the customer journey is based on omni-channel interactions that may last hours or even days.

Customer journeys can vary in length, scope and complexity. A given customer journey typically spans more than one customer interaction. Customer journeys beyond a single transaction are often considered relationship-building customer journeys ([Figure 6.10](#)). A primary objective of digital transformation initiatives is to make customer journeys as customer-centric as possible so that the nature of the relationship-building that occurs leads to positive, recurring customer journeys.

Figure 6.10

A customer journey involving interactions with a car dealership over a period of days in order to purchase a new car.

customer journey

1



Customer visits car dealership to view cars and orders a new yellow car.

2



Customer completes online credit application to apply for financing.

3



Dealership calls customer to confirm that the financing has been approved and that the car has been ordered.

Customer Data Intelligence

A fundamental means by which digital transformation solutions enable and realize customer-centricity is through the extensive collection, processing and analysis of customer-related data.

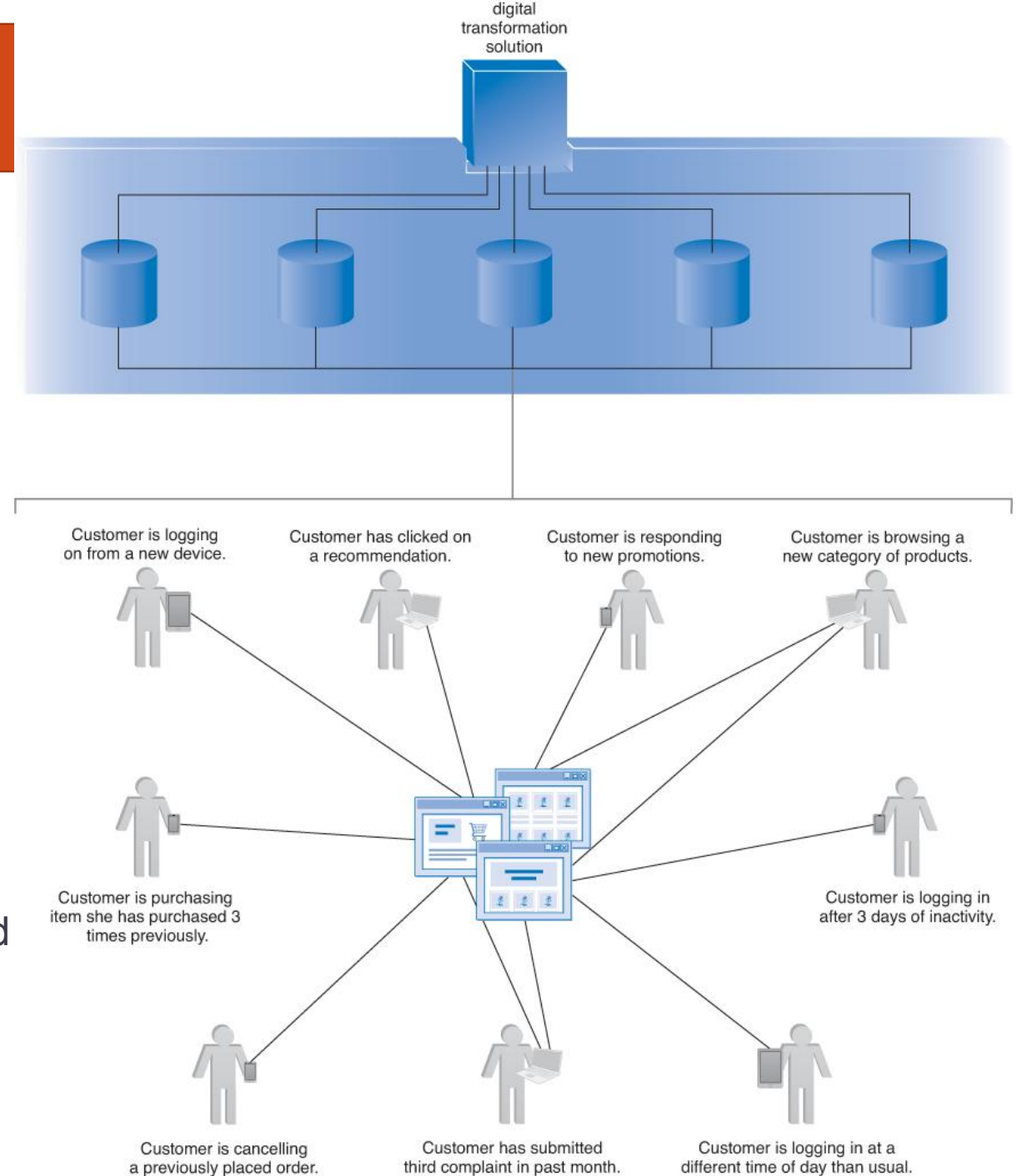
Examples of the types of customer-related data that can be collected include:

- current contact information
- prior contact information
- geographic location while logging in
- shipping destinations
- shipping speed preferences
- device being used for online access
- interaction patterns
- duration of stay (online session)
- hobbies and interests
- spending behavior
- past purchases
- prior returns
- tone of prior communication (friendly, unfriendly, abusive, etc.)
- relationships and friends with others (customer or non-customer)
- prior reviews and ratings left online

This type of data is constantly collected in detail and stored in the digital transformation environment ([Figure 6.11](#)).

Figure 6.11

A digital transformation solution provides an online application designed to capture and track a range of specific customer-related events and behaviors. Each is stored in a repository where it may be used to create historic customer profiles and also provide input for various analysis and analytics reports to help the organization better identify common customer trends and preferences.





Customer-centric data collected over time helps organizations build profiles of customers by analyzing the accumulated data to derive statistical profiles, such as:

- personality profile (likes, dislikes, preferences, social status, social preferences, etc.)
- session profile (average duration of stay, quantity and types of items viewed, etc.)
- spending profile (average amount spent per visit/week/year, spending habits, etc.)
- relationship profile (responsiveness to promotions, price reductions, recommendations, etc.)
- communication profile (tone and nature of messages, reviews, ratings, etc.)

This type of profile information can be fed into the online solution in realtime to enable it to continually enhance interactions during the next customer visit by customizing its relationship-value actions to the customer profiles. It also enables the solution to take anticipatory actions based on predictions it can derive from analyzing the accumulated customer data.

By combining profile insights with predictions, the solution can determine how to best guide and interact with the customer. For example, it may have logic that determines probabilities and then takes a course of action, as follows:

There is a 78% probability the customer will purchase Products A and B during this visit. Based on the customer's past preferences, there is a 57% probability the customer will be interested in this new Product C. Let's offer the customer a promotion whereby Product C is offered at a 20% discount when purchased together with Products A and B.

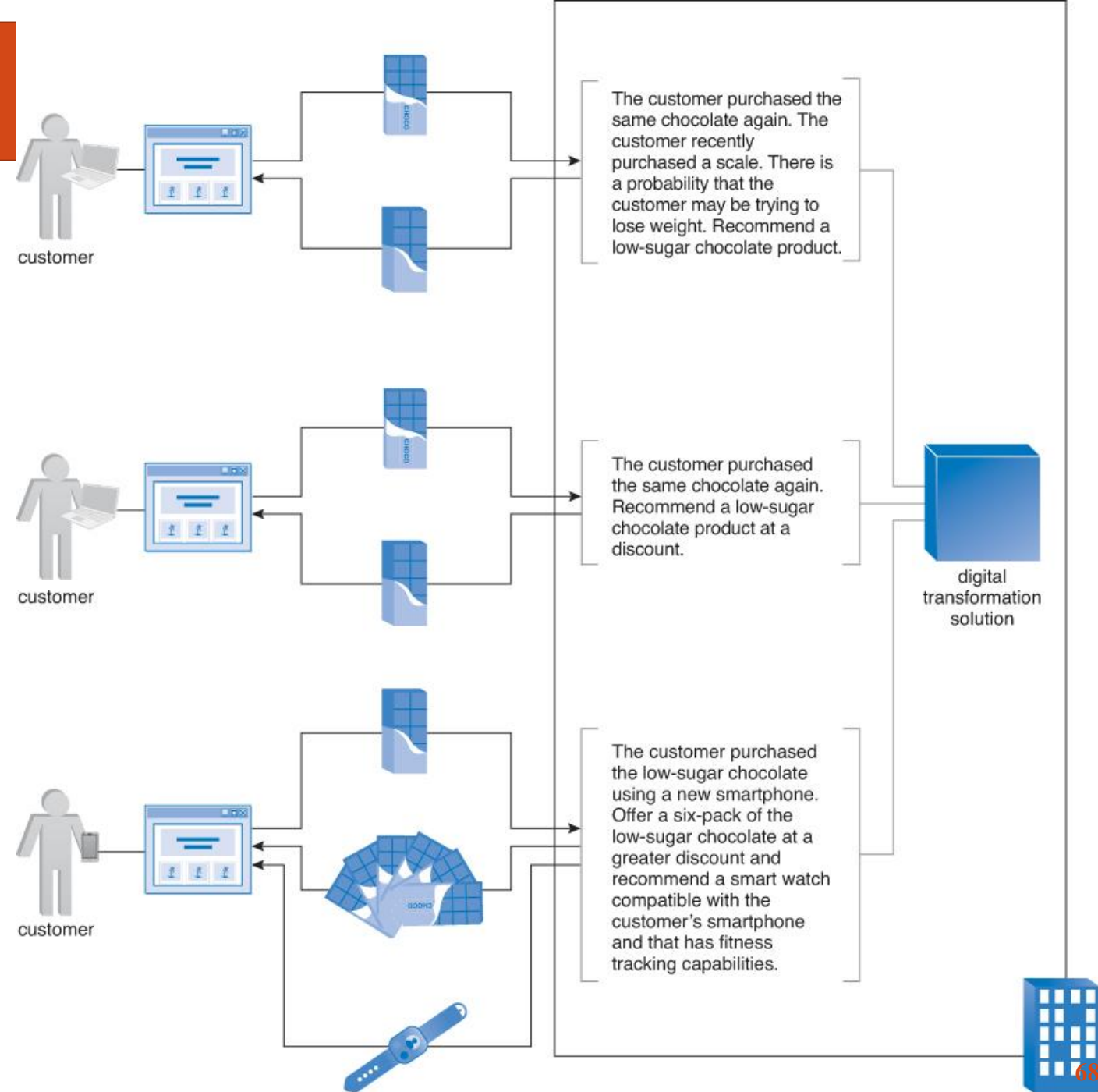
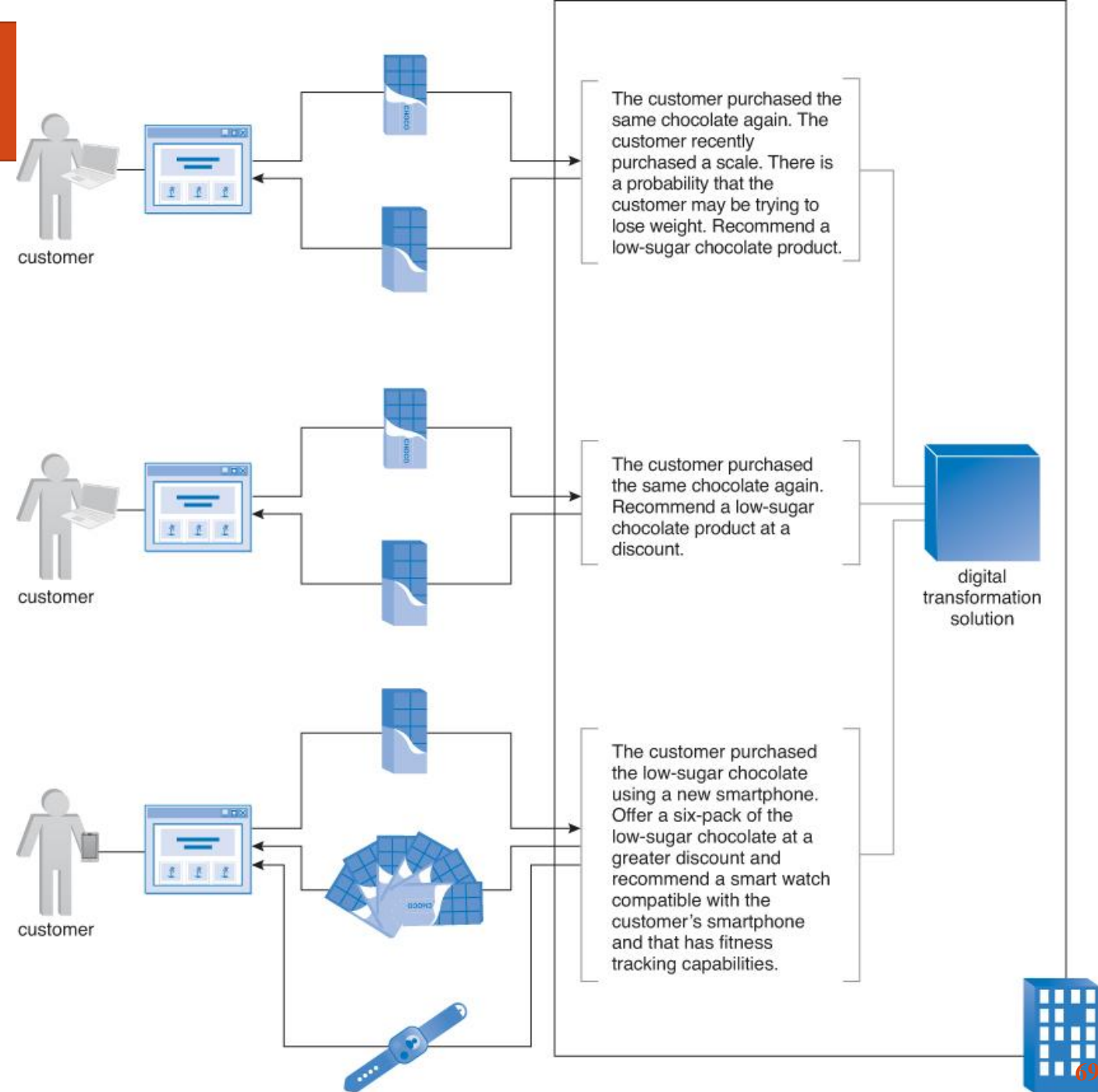




Figure 6.12
As a customer's purchasing profile and behavior changes, the digital transformation solution continuously adapts.



In these cases, traditional self-service portals provide relatively simple, transaction-value features that lead to a transaction-centric customer journey. By revisiting the underlying business processes, opportunities to introduce relationship-value steps can be found in an attempt to make the customer journey more customer-centric, as shown in [Figure 6.13](#).

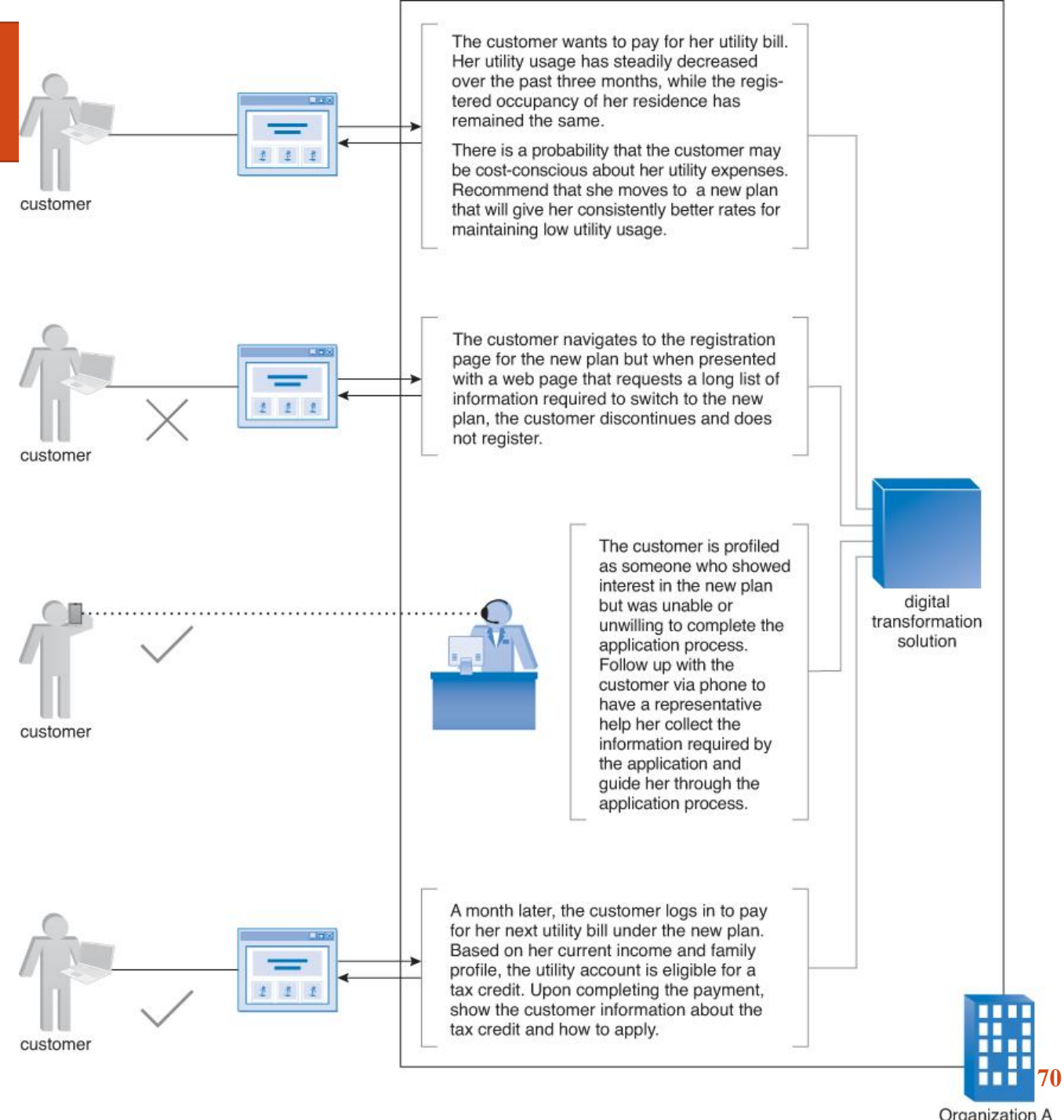


Figure 6.13

A customer is proactively guided to improve their relationship with a utility services organization.

